CHAPTER 1

How many primes are there?

Once one knows that there are infinitely many primes, one might ask how many primes there are up to 100, or up to 1000, or up to any given point x. Calculations reveal that the primes seem to make up a decreasing proportion of the positive integers. Can we explain this? And can we give a formula for how many primes there are up to x? Or at least give a good estimate?

1.1. The Prime Number Theorem

When Gauss was a teenager he was presented with a table of natural logarithms,[1] and then a later supplement which contained a table of primes up to 4000. This data made Gauss curious as to whether he could accurately guess the number of primes up to a given x, and he had a surprising idea: There are 168 primes up to 1000, 135 between 1000 and 2000, then 127 and 120 in the next intervals of length a thousand. Gauss observed that, for the natural logarithm (that is, to base e),

$$168 \cdot \log(10^3) \approx 1160, \quad 135 \cdot \log(2 \cdot 10^3) \approx 1026, \quad 127 \cdot \log(3 \cdot 10^3) \approx 1017;$$

in other words, he multiplied the number of primes in an interval by the logarithm of the size of those primes and each time seemed to get a value close to 1000. Subsequently the values of $\#\{\text{primes in } (x \cdot 10^3, (x+1) \cdot 10^3)\} \times \log((x+1) \cdot 10^3)$ are

$$995, \quad 1014, \quad 992, \quad 1036, \quad 962, \quad 1002, \quad 1032, \dots$$

which are arguably converging to 10^3. Years later he explained his findings:

> *In 1792 or 1793 ... I turned my attention to the decreasing frequency of primes ... counting the primes in intervals of length* 1000*. I soon recognized that behind all of the fluctuations, this frequency is on average inversely proportional to the logarithm. ...*
>
> — from a letter to Encke by K. F. Gauss (Christmas Eve 1849)

In 1808 Legendre had already quantified how quickly $\pi(x)/x$ diminishes, suggesting that

$$\pi(x) \Big/ \frac{x}{\log x} \to 1 \quad \text{as } x \to \infty. \tag{1.1}$$

This follows from Gauss's observations, but more, they suggest that

About 1 in $\log x$ *of the integers near* x *are prime,*

[1] Before the advent of electronic calculators and computers, logarithms provided a fast way to accurately multiply, using the formula $\log a + \log b = \log ab$, as it is easy to add numbers quickly. To do this efficiently by hand we need a way to determine $\log a$ from a, $\log b$ from b, and then ab from $\log ab$. This can be quickly done with a look-up table for logarithms.

which is (subtly) different from Legendre's assertion: Gauss's observation suggests that a good approximation to the number of primes up to x is $\sum_{n=2}^{x} \frac{1}{\log n}$. As $\frac{1}{\log t}$ does not vary much for t between n and $n+1$, Gauss deduced that $\pi(x)$, the number of primes up to x, should be well approximated by

$$\int_2^x \frac{dt}{\log t}. \tag{1.2}$$

We denote this quantity by $\mathrm{Li}(x)$ and call it *the logarithmic integral*.[2] Let's compare this to the data for Gauss's intervals of length 1000:

$(m \cdot 10^3, (m+1)\cdot 10^3)$	$m=0$	1	2	3	4	5	6	7	8	9
Number of primes	168	135	127	120	119	114	117	107	110	112
Gauss's prediction	178	137	128	123	119	116	114	112	111	109

FIGURE 1.1. Number of primes in $(m \cdot 10^3, (m+1)\cdot 10^3)$ vs. Gauss's $\int\limits_{m\cdot 10^3}^{(m+1)\cdot 10^3} \frac{dt}{\log t}$.

What a remarkably good fit! Next we compare Gauss's prediction with the actual count of primes up to various values of x:

x	$\pi(x) := \#\{\text{primes} \le x\}$	Overcount: $\mathrm{Li}(x) - \pi(x)$
10^3	168	10
10^4	1229	17
10^5	9592	38
10^6	78498	130
10^7	664579	339
10^8	5761455	754
10^9	50847534	1701
10^{10}	455052511	3104
10^{11}	4118054813	11588
10^{12}	37607912018	38263
10^{13}	346065536839	108971
10^{14}	3204941750802	314890
10^{15}	29844570422669	1052619
10^{16}	279238341033925	3214632
10^{17}	2623557157654233	7956589
10^{18}	24739954287740860	21949555
10^{19}	234057667276344607	99877775
10^{20}	2220819602560918840	222744644
10^{21}	21127269486018731928	597394254
10^{22}	201467286689315906290	1932355208

TABLE 1.1. Primes up to various $x \le 10^{29}$, and the overcount in Gauss's prediction. (Table 1.1 continues on next page.)

[2]Some authors begin the integral defining $\mathrm{Li}(x)$ at $x = 0$. This adds a complication since the integrand equals ∞ at $x = 1$; nonetheless this can be handled, and the difference between the two definitions is then the constant $1.045163\ldots$, which has little relevance to the discussion here.

x	$\pi(x) := \#\{\text{primes} \leq x\}$	Overcount: $\text{Li}(x) - \pi(x)$
10^{23}	1925320391606803968923	7250186216
10^{24}	18435599767349200867866	17146907278
10^{25}	176846309399143769411680	55160980939
10^{26}	1699246750872437141327603	155891678121
10^{27}	16352460426841680446427399	508666658006
10^{28}	157589269275973410412739598	1427745660374
10^{29}	1520698109714272166094258063	4551193622464

TABLE 1.1 (continued). Primes up to various $x \leq 10^{29}$, and the overcount in Gauss's prediction.

Gauss's prediction is amazingly accurate. From the known data, Gauss's prediction $\text{Li}(x)$ seems to be larger than $\pi(x)$ for each $x \geq 8$, though only by a small amount.[3] To quantify this "small amount", we observe that the last column (representing the overcount) seems to always be about half the width of the central column (representing the number of primes up to x), so this data suggests that the difference is no bigger than a small multiple of $\sqrt{x}$. More precisely, does there perhaps exist a constant $c > 0$ such that[4]

$$|\pi(x) - \text{Li}(x)| < c\sqrt{x},$$

for all $x \geq 2$? This might be optimistic but, at the very least, the ratio of the number of primes $\pi(x)$ to Gauss's guess, $\text{Li}(x)$, should tend to 1 as $x \to \infty$; that is,

$$\pi(x)/\text{Li}(x) \to 1 \quad \text{as } x \to \infty.$$

Now $\text{Li}(x)/\frac{x}{\log x} \to 1$ as $x \to \infty$.[5] Combining these last two limits, we deduce (1.1). The notation of limits is cumbersome; it is easier to write

$$\boxed{\pi(x) \sim \frac{x}{\log x}} \tag{1.3}$$

(which means the same thing) and we say that "$\pi(x)$ *is asymptotic to* $x/\log x$".[6]

The asymptotic (1.3) is called *the Prime Number Theorem.* It was first proved in 1896 (by Hadamard and de la Vallée Poussin independently), more than 100 years

[3] The data is misleading in that $\text{Li}(x)$ is not larger than $\pi(x)$ for all x, but the first counterexample is far beyond where we can hope to calculate (see section 34.3 then section 34.5).

[4] We believe that the size of $\pi(x)$ is roughly $\frac{x}{\log x}$, whereas the error in this approximation is something like $\sqrt{x}$, which is smaller than the main term by a factor of roughly $\sqrt{x}$ — we call this *square root cancelation*, a phenomenon we will see a lot more of.

[5] As may be proved using L'Hôpital's rule. We will discuss this further in section 1.3.

[6] One goal of the first few chapters is to help the reader become familiar with the notation of analytic number theory. Not just interpreting, but also manipulating it, as this will be taken for granted in what we do. Here we introduce $A(x) \sim B(x)$; that is, $A(x)$ is asymptotic to $B(x)$, which is equivalent to $\lim_{x\to\infty} A(x)/B(x) = 1$. It does *not* mean that "$A(x)$ is approximately equal to $B(x)$", which has no strict mathematical meaning. Rather it means that for all $\epsilon > 0$, no matter how small, one has

$$(1-\epsilon)B(x) < A(x) < (1+\epsilon)B(x)$$

once x is sufficiently large (where how large is " sufficiently large" depends on ϵ). This definition concerns the ratio $A(x)/B(x)$, *not* their difference $A(x) - B(x)$. Therefore $x - 1 \sim x$ and $x + x/\log x \sim x$ are equally true, even though the first is a better approximation to x than the second.

(Dan Shanks, 1985)

after Gauss's guess. The Prime Number Theorem is a high point of nineteenth-century mathematics, involving several remarkable developments. An important goal of this book is to sketch the original proof of the Prime Number Theorem, following an extraordinary program of ideas outlined by Riemann in 1859. This remains the best way to introduce the ideas of analytic number theory.[7]

It is difficult to prove the Prime Number Theorem, arguably because primes are defined in terms of what they are not (that is, they are positive integers that *are not* the product of two smaller positive integers), rather than in terms of what they are. Most direct descriptions of prime numbers are a bit convoluted, but the most useful description comes from the Riemann zeta-function.

Exercise 1.1.* Assume the Prime Number Theorem.

(a) Prove that for every fixed $c > 0$ we have $\pi(cx) \sim c\,\pi(x)$ as $x \to \infty$.
(b) Show that for all $\epsilon > 0$, if x is sufficiently large, then there are primes between x and $x + \epsilon x$.
(c) Deduce that $\mathbb{R}_{\geq 0}$ is the set of limit points of the set $\{p/q : \ p, q \text{ primes}\}$.

Exercise 1.2.* Assume the Prime Number Theorem.

(a) Show that there are infinitely many primes whose leading digit is a "1". How about leading digit "7"?
(b) Let $a_1, \ldots, a_d$ be any sequence of digits, that is, integers between 0 and 9, with $a_1 \neq 0$. Show that there are infinitely many primes whose d leading digits are $a_1, \ldots, a_d$.

The frequency of these digit patterns will be discussed further in section 1.4.

Exercise 1.3. Let $p_1 = 2 < p_2 = 3 < \cdots$ be the sequence of primes.

(a) Assume the Prime Number Theorem and prove that

$$p_n \sim n \ \log n \quad \text{as } n \to \infty.$$

Explicitly it can be shown that if $n \geq 6$, then

$$n(\log n + \log\log n - 1) < p_n < n(\log n + \log\log n).$$

(b) Conversely show that if $p_n \sim n \ \log n$, then the Prime Number Theorem holds.

Exercise 1.4. Assume the Prime Number Theorem. Show that the sum of all of the primes $\leq x$ is $\sim x^2/(2\log x)$.

1.2. The first appearance of the Riemann zeta-function

We define $\zeta(s)$, the *Riemann zeta-function* by

$$\zeta(s) := \sum_{\substack{n\geq 1 \\ n \text{ a positive integer}}} \frac{1}{n^s} \qquad \text{whenever } s \text{ is a complex number with } \operatorname{Re}(s) > 1.$$

Here the sum is over all positive integers n, summed in ascending order.[8]

[7]In Part 1 we will introduce some of the more elementary concepts and in Part 4 the techniques of analytic number theory. Explaining these basics will allow us to explore different aspects of the distribution of prime numbers in subsequent parts of this book. In particular we will be ready to sketch the proof of the Prime Number Theorem in Part 5.

[8]Indeed, throughout, we assume that any infinite sum is summed over the indexing variable in "ascending order", unless specified otherwise.

Now if $s = \sigma + it$ with $\sigma > 1$ and $t \in \mathbb{R}$, then this sum is *absolutely convergent* since the sum of the absolute values of the terms is

$$\sum_{n\ge 1}\left|\frac{1}{n^s}\right| = \sum_{n\ge 1}\frac{1}{n^\sigma} \le 1 + \int_1^\infty \frac{dt}{t^\sigma} = 1 + \left[\frac{t^{1-\sigma}}{1-\sigma}\right]_1^\infty = 1 + \frac{1}{\sigma-1} = \frac{\sigma}{\sigma-1}. \tag{1.4}$$

Here we have used that $n^{-\sigma} < \int_{n-1}^n t^{-\sigma} dt$ for each $n \ge 2$, which holds since $t^{-\sigma}$ is a decreasing function in t.[9]

In 1737, Euler showed that if s is a real number > 1, then

$$\sum_{\substack{n\ge 1 \\ n \text{ a positive integer}}} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1}. \tag{1.5}$$

To prove (1.5) (not only for real $s > 1$ but also for all complex s with $\mathrm{Re}(s) > 1$), we use the Fundamental Theorem of Arithmetic which states that every positive integer $n \ge 1$ can be written in a unique way as a product of primes. This implies that if $p_1 = 2 < p_2 = 3 < \cdots$ is the sequence of primes, then the set $\{n \ge 1 : n \text{ is a positive integer}\}$ is precisely given by the set

$$\{p_1^{a_1} p_2^{a_2} \cdots : \ a_1, a_2, \ldots \ge 0 \text{ with only finitely many } a_k \ne 0\}.$$

Therefore an absolutely convergent sum involving the elements of the positive integers equals the analogous sum involving the factorizations given in the set in the last displayed equation. In particular,

$$\begin{aligned}\sum_{\substack{n\ge 1 \\ n \text{ a positive integer}}} \frac{1}{n^s} &= \sum_{a_1, a_2, \ldots, a_k \ge 0} \frac{1}{(p_1^{a_1} p_2^{a_2} \cdots)^s} \\ &= \left(\sum_{a_1\ge 0} \frac{1}{(p_1^{a_1})^s}\right)\left(\sum_{a_2\ge 0} \frac{1}{(p_2^{a_2})^s}\right)\cdots .\end{aligned}$$

This rearrangement is valid as the sum of everything here is absolutely convergent; and (1.5) follows as each of the sums in the product on the right-hand side is over a geometric progression, and summing them all simultaneously is valid as both sides of the equation converge absolutely.

A product $\prod_{j\ge 1}(1 - z_j)$ with each $|z_j| < 1$ converges absolutely if $\sum_{j\ge 1} |z_j|$ converges (in which case the product converges to a non-zero value). As each $|p^{-s}| = |p^{-\sigma}| < p^{-1} < 1$, we deduce that the product in (1.5), known as the *Euler product*, converges absolutely when $\mathrm{Re}(s) > 1$ as

$$\sum_{p \text{ prime}} \left|\frac{1}{p^s}\right| = \sum_{p \text{ prime}} \frac{1}{p^\sigma} \le \sum_{n\ge 1} \frac{1}{n^\sigma} \le \frac{\sigma}{\sigma - 1},$$

the final inequality following from (1.4). The convergence of the Euler product in (1.5), together with none of the individual terms in the product equalling 0, implies that the Euler product cannot equal 0; therefore,

$$\zeta(s) \ne 0 \quad \text{for all } s \text{ with } \mathrm{Re}(s) > 1. \tag{1.6}$$

[9] In this proof we have written $\sum_{n\ge 1}$ in place of $\sum_{\substack{n\ge 1 \\ n \text{ a positive integer}}}$ for simplicity, and we will continue to abbreviate the subscripts of sums like this when their full meaning should be clear from the context.

Euler was the first to see how (1.5) could be used to begin to understand the number of primes. He deduced from (1.5) that

$$\sum_{p \text{ prime}} \frac{1}{p} \text{ diverges,} \tag{1.7}$$

suggesting how numerous the primes are in comparison to other sequences of integers. For example, $\sum_{n\geq 1} \frac{1}{n^2}$ converges, so the primes are, in this sense, more numerous than the squares. This implies that there are arbitrarily large values of x for which $\pi(x) > \sqrt{x}$.

Proof of (1.7). Suppose not, so that $\sum_p \frac{1}{p}$ converges. Then the right-hand side of (1.5) converges to a non-zero constant as $s = \sigma \to 1^+$ (where $\sigma \in \mathbb{R}$), and therefore so does the left-hand side of (1.5), $\sum_{n\geq 1} \frac{1}{n^\sigma}$. However, if $\sigma > 1$, then $\frac{1}{n^\sigma} > \int_n^{n+1} \frac{dt}{t^\sigma}$ for each $n \geq 1$ as $1/t^\sigma$ is a decreasing function, and so

$$\sum_{n\geq 1} \frac{1}{n^\sigma} \geq \int_1^\infty \frac{dt}{t^\sigma} = \frac{1}{\sigma - 1}.$$

The right-hand side diverges to ∞ as $\sigma \to 1^+$, and therefore so does the left-hand side, a contradiction. □

The sum in (1.5) is absolutely convergent only when $\mathrm{Re}(s) > 1$ and so gives an unambiguously defined function in that range, but not for other values of s. Fortunately Riemann observed that the theory of analytic continuation comes to our rescue, giving a unique, useful way to extend $\zeta(s)$ to a function that is well-defined on the whole complex plane (we will discuss these notions in chapter 18); for this reason it is known as the *Riemann zeta-function*.

We will see that the Prime Number Theorem (as stated in (1.3)) is equivalent to the seemingly unrelated statement that the analytic continuation of $\zeta(s)$ has no zeros ρ with $\mathrm{Re}(\rho) = 1$ (as proved in chapter 23).[10] It is surprising that an elementary statement about primes can be equivalent to a statement about the zeros of an analytic continuation, an intrinsically non-elementary concept, and it is perhaps for this reason that every known proof of the Prime Number Theorem is substantial (see appendix D).

Exercise 1.5. Prove that if $\sigma > 1$, then $0 < \max\{1, \frac{1}{\sigma-1}\} < \zeta(\sigma) < \frac{\sigma}{\sigma-1}$.

Exercise 1.6.* By looking at Euler products, prove that if $\sigma > 1$ and $s = \sigma + it$, then

$$\frac{\sigma - 1}{\sigma} < \frac{\zeta(2\sigma)}{\zeta(\sigma)} \leq |\zeta(s)| \leq \zeta(\sigma) < \frac{\sigma}{\sigma - 1}.$$

Exercise 1.7. Use (1.7) to deduce that there are arbitrarily large x for which $\pi(x) > x/(\log x)^2$.

The rest of this chapter is a little more technical, taking a first look at some of the specialized notation and techniques of analytic number theory.

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[10] "Equivalent" in the sense that each claim easily implies the other.

1.3. Better understanding Li(x), and quantitative notation

Gauss's heuristic suggests that $\pi(x)$ is well approximated by the complicated function $\mathrm{Li}(x)$.[11] To better understand $\mathrm{Li}(x)$, we integrate by parts

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t} = \frac{x}{\log x} - \frac{2}{\log 2} + \int_2^x \frac{dt}{(\log t)^2}. \tag{1.8}$$

To confirm that the "main term" is $\frac{x}{\log x}$, we need to show that the other terms on the right-hand side are significantly smaller as $x \to \infty$. The constant $\frac{2}{\log 2}$ is bounded and so smaller, and the integral is $\sim \frac{x}{(\log x)^2}$ by L'Hôpital's rule. Another way to bound the integral is to note that $\frac{1}{(\log t)^2} \le \frac{1}{(\log 2)^2}$ if $2 \le t \le \sqrt{x}$, and $\frac{1}{(\log t)^2} \le \frac{1}{(\log \sqrt{x})^2}$ if $\sqrt{x} \le t \le x$, so that

$$\int_2^x \frac{dt}{(\log t)^2} \le \int_2^{\sqrt{x}} \frac{dt}{(\log 2)^2} + \int_{\sqrt{x}}^x \frac{4dt}{(\log x)^2} < c\sqrt{x} + \frac{4x}{(\log x)^2},$$

where $c = 1/(\log 2)^2$. Now $\sqrt{x}$ grows a lot more slowly than $x/(\log x)^2$ as $x \to \infty$, and so $c\sqrt{x} \le x/(\log x)^2$ if x is sufficiently large, which implies that

$$\int_2^x \frac{dt}{(\log t)^2} \le \frac{5x}{(\log x)^2}.$$

We want to focus here on the function $x/(\log x)^2$ of x, not the constant "5" in front which was derived fairly arbitrarily.[12] In such arguments it is best not to aim for a particular constant and so we have notation which ignores the specific value. We write

$$\mathrm{Li}(x) = \frac{x}{\log x} + O_{x\to\infty}\left(\frac{x}{(\log x)^2}\right).$$

The notation $O_{x\to\infty}(g(x))$ represents a function $f(x)$ for which there exists a constant $c > 0$ such that $|f(x)| \le c\,g(x)$ once x is sufficiently large. If there is only one variable that could be going to ∞, then we abbreviate this to $O(g(x))$. The notation does not specify the value of the constant c; we only know that we may take its value to be some positive real number. We say that "$f(x)$ is big Oh of $g(x)$". In our example we could have taken $c = 5$ but, as we have seen, the precise value of c is irrelevant.

Another useful notation is $f(x) = o_{x\to\infty}(g(x))$ which means that $f(x)/g(x) \to 0$ as $x \to \infty$; we say that "$f(x)$ is little Oh of $g(x)$". Again we may abbreviate this to $f(x) = o(g(x))$ if there is no mistaking which variable goes to ∞. As an example we note that $\frac{x}{(\log x)^2} = o(\frac{x}{\log x})$ and so the last displayed equation implies that

$$\mathrm{Li}(x) = \frac{x}{\log x} + o\left(\frac{x}{\log x}\right), \text{ which can also be rewritten as}$$

$$\mathrm{Li}(x) = \{1 + o(1)\}\frac{x}{\log x}, \text{ or even as } \mathrm{Li}(x) \sim \frac{x}{\log x},$$

as follows from exercise 1.9.

Exercise 1.8. Prove that $r(x) \to L$ as $x \to \infty$ if and only if $r(x) = L + o_{x\to\infty}(1)$.

[11] "Complicated" in the sense that it is defined by an integral that cannot be evaluated in "closed form" as a finite expression involving more familiar-looking functions.

[12] That is, minor variants of the same argument would lead to a different constant. For example, by partitioning the integral at $x^{3/4}$ one can obtain the bound $\le 2x/(\log x)^2$ once x is sufficiently large.

Exercise 1.9. Prove that $f(x) \sim g(x)$ if and only if $f(x) = \{1 + o(1)\}g(x)$.

Exercise 1.10. Decide which of the following are true or false, and explain why. (a) $O(x) - O(x) = O(x)$; (b) $O_{x\to\infty}(x) = o_{x\to\infty}(x^2)$; (c) $O_{x\to 0}(x) = O_{x\to 0}(x^2)$; (d) $O_{x\to\infty}(x^2) = O_{x\to\infty}(x)$; and (e) $O_{x\to 0}(x^2) = o_{x\to 0}(x)$.

Exercise 1.11. Prove that $\sum_{n=2}^{N} \frac{1}{\log n} = \mathrm{Li}(N) + O(1)$.

If we integrate the expression for $\mathrm{Li}(x)$ m times by parts (starting as in (1.8)), then we obtain the formula

$$\mathrm{Li}(x) = \sum_{k=0}^{m-1} \frac{k!x}{(\log x)^{k+1}} + O_m(1) + m! \int_2^x \frac{dt}{(\log t)^{m+1}}.$$

The constant in the big Oh depends only on m, which is why it is given as a subscript. The last two terms on the right-hand side are $O(\frac{m!x}{(\log x)^{m+1}})$ and so

$$\mathrm{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + \cdots + \frac{(m-1)!\,x}{(\log x)^m} + O\left(\frac{m!x}{(\log x)^{m+1}}\right).$$

One might guess that we could extend this sum to the infinite series

$$\frac{x}{\log x} + \frac{x}{(\log x)^2} + \frac{2x}{(\log x)^3} + \cdots + \frac{(m-1)!\,x}{(\log x)^m} + \cdots$$

but this diverges for every x. This is why the truncation at m terms provides a useful approximation for $\mathrm{Li}(x)$, but one cannot easily extend this.

Exercise 1.12.*

(a) Prove that $\mathrm{Li}(x) = \frac{x}{\log x} + \{1 + o(1)\}\frac{x}{(\log x)^2}$.
(b) Deduce that $\mathrm{Li}(x) > \frac{x}{\log x}$ if x is sufficiently large.
(c) Prove that $\frac{1}{\log 2x} = \frac{1}{\log x} - \frac{\log 2}{(\log x)^2} + O\left(\frac{1}{(\log x)^3}\right)$.
(d) Deduce that $\mathrm{Li}(2x) - \mathrm{Li}(x) < \mathrm{Li}(x)$ if x is sufficiently large. (See a nice application of this estimate in exercise 36.1(a).)

Exercise 1.13. After finishing exercise 1.12 prove the following:

(a) Fix $\epsilon > 0$. Prove that if x is sufficiently large, then $\frac{x}{\log x - 1} < \mathrm{Li}(x) < \frac{x}{\log x - (1+\epsilon)}$.
(b) Explain why 1 is the "best" choice for B when approximating $\mathrm{Li}(x)$ by $x/(\log x - B)$.
(c) Prove that $\mathrm{Li}(x) = \frac{x}{\log x - 1} + \{1 + o(1)\}\frac{x}{(\log x)^3}$.

Exercise 1.14.* Prove that

$$\sum_{n=3}^{N} \left(\frac{1}{\log n} - \frac{1}{(\log n)^2}\right) = \frac{N}{\log N} + O(1).$$

1.4. Notions of density

A set of positive integers A has *density* (or *natural density*) δ if

$$\#\{n \le x : n \in A\} \sim \delta x \quad \text{as } x \to \infty.$$

There are many natural examples; for example the integers with a given final digit have natural density $\frac{1}{10}$, since, for instance

$$\#\{n \le x : n \equiv 7 \pmod{10}\} = \frac{x}{10} + O(1).$$

However, there are other natural sets that do not have a natural density:

Exercise 1.15.

(a) Show that for any integer $a, 1 \leq a \leq 9$, there are $\sim x/9$ integers $\leq x$ whose leading digit is a, where $x = 10^n$, and integer $n \to \infty$.

(b) Show that there are $\sim 5x/9$ integers $\leq x$ whose leading digit is 1, where $x = 2 \cdot 10^n$, and integer $n \to \infty$.

Therefore the proportion of integers whose leading digit is 1 veers between $\frac{1}{9}$ and $\frac{5}{9}$ and so has no limit. Therefore we need another way to measure the "density" of this set.

Exercise 1.16.* For any integer $a \in \{1, \ldots, 9\}$ let $\mathcal{N}_a$ denote the set of positive integers with leading digit a. Prove that

$$\lim_{x\to\infty} \frac{1}{\log x} \sum_{n\leq x,\ n\in\mathcal{N}_a} \frac{1}{n} \text{ exists and equals } \frac{\log(1 + 1/a)}{\log 10}.$$

This is the *logarithmic density* of the set $\mathcal{N}_a$.

Exercise 1.17. What can you say about the density of the integers whose second digit is 0?

A similar thing happens with the primes:

Exercise 1.18.* Assume the Prime Number Theorem.

(a) Show that the proportion of primes whose leading digit is 1 veers between $\frac{1}{9}$ and $\frac{5}{9}$.

(b) Establish, with proof, an appropriate notion of density for the primes with leading digit 1.

(c) Establish an analogous result when the leading k digits are given.

1.5. How good an approximation is $\mathrm{Li}(x)$?

The Prime Number Theorem is the assertion that $\pi(x)-\mathrm{Li}(x) = o(\frac{x}{\log x})$ though the data in Table 1.1 suggests that the difference is much smaller, perhaps as small as $O(x^{1/2})$, so it is interesting to appreciate what can be proved unconditionally. Even the weaker upper bound $O(x^B)$ for some $B \in [\frac{1}{2}, 1)$ is beyond what we can currently prove unconditionally, but we can unconditionally "win" by any power of $\log x$: That is, fix $A > 1$ and then

$$\pi(x) - \mathrm{Li}(x) = O_A\left(\frac{x}{(\log x)^A}\right).$$

In our more advanced book [**A-6**, Theorem 23.1] we prove that

$$\pi(x) - \mathrm{Li}(x) = O\big(x \exp\big(-\sqrt{\log x}\big)\big)$$

which is considerably better, first proved in 1899 by de la Vallée Poussin. In the best bounds known one replaces the $\sqrt{\log x}$ here with $\frac{c(\log x)^{3/5}}{(\log\log x)^{1/5}}$ for some constant $c > 0$. This was proved by Vinogradov and Korobov in the 1950s and has not been improved since so, presumably, it will take a fundamentally new idea to do better.

Questions on notation

To study analytic number theory one needs to appreciate and manipulate the notation and concepts to do with estimates. The following exercises will at first be challenging but, after some effort, second nature to the reader.

Exercise 1.19.

(a) Prove that $e^{o(1)} = 1 + o(1)$.
(b) Show that if $f(x) = 1 + O(\epsilon)$ for every $\epsilon > 0$, then $f(x) = 1 + o(1)$.[13]
(c) Show that if $A(x) \asymp B(x)$, then $\log A(x) = \log B(x) + O(1)$.[14]

Exercise 1.20. Prove that if $a, b > 0$, then $\max\{a, b\} \asymp a + b$.

Exercise 1.21. Fix $r \geq 0$ and $\epsilon > 0$.

(a) Prove that $t^r \leq e^t$ for t sufficiently large.
(b) Deduce that $t^r = O_r(e^t)$ for all $t \geq 0$.
(c) Deduce that $(\log x)^r = O_r(x)$ for all $x \geq 1$ and that $\log n = O_\epsilon(n^\epsilon)$ for all $n \geq 1$.

Exercise 1.22. Prove that if $y^2 = x^2 + O_{x\to 0}(x^3)$ with $x, y > 0$, then $y = x + O_{x\to 0}(x^2)$.

Exercise 1.23. Sort the following functions in ascending order, as $x \to \infty$:

$$(\log x)^5,\ \exp(\sqrt{\log x}),\ x^{1/3},\ e^{\sqrt{x \log x}},\ \exp((\log\log x)^2),\ \exp(\sqrt{\log x \log\log x}).$$

Exercise 1.24. Let $A(x) = x + \sqrt{x}$. Show that $\{1 + o(1)\}A(x) - \{1 + o(1)\}x$ is not necessarily equal to $\{1 + o(1)\}\sqrt{x}$, and indicate what we can deduce that it is equal to.

Exercise 1.25.* Assume that for all $\epsilon > 0$ there exists a constant $\kappa(\epsilon)$ for which $|a_n| \leq \kappa(\epsilon)n^\epsilon$. Prove that $a_n = n^{o(1)}$ as $n \to \infty$.[15]

Exercise 1.26. Assume that $a_n = n^{o(1)}$ for all integers $n \geq 1$.

(a) Prove that for all $\epsilon > 0$ there exists a constant $\kappa(\epsilon)$ for which $|a_n| \leq \kappa(\epsilon)n^\epsilon$.
(b)* Prove that if $\mathrm{Re}(s) > 1$, then the series $\sum_{n\geq 1} a_n/n^s$ is absolutely convergent.

Exercise 1.27.† Fix $\beta \in \mathbb{R}$. Prove that $\sum_{n\leq x} |a_n| \leq x^{\beta+o(1)}$ if and only if $\sum_{n\geq 1} a_n/n^s$ is absolutely convergent for all s with $\mathrm{Re}(s) > \beta$.

[13] Remember that $g(x) = O(\epsilon)$ means that there exists a constant $C > 0$ (independent of any ϵ) such that for any $\epsilon > 0$ there exists x_ϵ such that if $x \geq x_\epsilon$, then $|g(x)| \leq C\epsilon$. We call C an *absolute constant* as it does not depend on any variable, like ϵ. For example if $g(x) = 1/x$, then $|1/x| < \epsilon$ whenever $x > 1/\epsilon$.

[14] We remind the reader that we write $f(x) \asymp g(x)$ if $f(x) = O(g(x))$ and $g(x) = O(f(x))$.

[15] Taking logarithms, this is equivalent to showing the 1-sided inequality $\log|a_n| \leq o(\log n)$.

CHAPTER 3

Partial summation, and consequences of the Prime Number Theorem

In this chapter we will introduce the general technique of partial summation which allows us to easily go from one sum over primes to another. Under the assumption of the Prime Number Theorem, we will use this technique to estimate various sums over primes which will be useful later.

3.1. Partial summation

Given a sequence of complex numbers a_n and some continuous function $f : \mathbb{R} \to \mathbb{C}$, we wish to estimate the value of

$$\sum_{n=A+1}^{B} a_n f(n)$$

given that we already have good estimates for the partial sums

$$S(t) := \sum_{k \le t} a_k.$$

For example, if $a_n = 1$ when n is prime and $a_n = 0$ otherwise, then $S(x) = \pi(x)$, and if $f(t) = \log t$ with $A = 0$ and $B = x$, then the displayed sum is $\Theta(x)$.

In the technique of partial summation, the idea is that $S(t)$ changes value by a_n exactly when $t = n$; formally we might write "$dS(n) = a_n$" and $dS(t) = 0$ otherwise. Then we can "integrate" $f(t)$ from A to B with respect to $dS(t)$, and so[1]

$$\sum_{A<n\le B} a_n f(n) = \int_{A^+}^{B^+} f(t)d(S(t)).$$

Typically we evaluate this by integrating by parts.

We need to be rigorous: The general idea is to provide an appropriate meaning to $\int_a^b f(t)dg(t)$ where $g(t)$ might not be a continuous function. To do so we break the integral up into very short intervals, call them $[x_n, x_{n+1}]$, so that $f(t)$ is more or less constant on the interval when $g(t)$ changes by $g(x_{n+1}) - g(x_n)$. This inspires the definition of the *Riemann–Stieltjes integral*, $\int_a^b f(t)dg(t)$, given by

$$\lim_{\Delta\to 0} \sum_{n=0}^{N-1} f(c_n)(g(x_{n+1}) - g(x_n))$$

for any choice of the x_n with $a = x_0 < \cdots < x_N = b$ and $|x_{n+1} - x_n| \le \Delta$, where each $c_n \in [x_n, x_{n+1}]$.

[1]The notation "t^+" denotes a real number "marginally" larger than t, and "t^-" denotes a real number "marginally" smaller than t.

Typically f is continuously differentiable on $[A, B]$, so we can integrate by parts as follows:

$$\sum_{A<n\le B} a_n f(n) = \int_{A^+}^{B^+} f(t)d(S(t)) = [S(t)f(t)]_A^B - \int_A^B S(t)f'(t)dt$$
$$= S(B)f(B) - S(A)f(A) - \int_A^B S(t)f'(t)dt. \tag{3.1}$$

Moreover (3.1) holds for all non-negative real numbers $A < B$.[2] Evaluating the sum $\sum_{A<n\le B} a_n f(n)$ using an integral in this way is called *partial summation.*

For example, one useful consequence comes if $S(t) = M(t) + E(t)$ where $M(t)$, the main term, is differentiable, and $E(t)$ is the "error term", in which case

$$\sum_{A<n\le B} a_n f(n) = \int_{A^+}^{B^+} f(t)\frac{dM}{dt}dt + E(B)f(B) - E(A)f(A) - \int_A^B E(t)f'(t)dt. \tag{3.2}$$

For example, if we write $\pi(x) = \mathrm{Li}(x) + E(x)$, then

$$\sum_{p\le x} f(p) = \int_{2^-}^{x} \frac{f(t)}{\log t}dt + E(x)f(x) - E(2^-)f(2) - \int_2^x E(t)f'(t)dt. \tag{3.3}$$

It is important that we write $E(2^-)$ in the line above and not $E(2)$, since $E(x)$ is not continuous at $x = 2$ (or at any other prime).

3.2. Special constants and useful formulas

Let each $a_k = 1$ and $S(t) = \sum_{n\le t} 1 = [t] = t - \{t\}$ where $[t]$ is the *integer part* of t (the largest integer $\le t$) and $\{t\}$ $(= t - [t])$ is the *fractional part.* Therefore $M(t) = t$ and $E(t) = -\{t\}$ in the notation above. Taking $f(n) = \frac{1}{n}$ in (3.2) with N an integer, yields

$$\sum_{n=1}^{N} \frac{1}{n} = \int_{1^-}^{N} \frac{d[t]}{t} = \int_{1^-}^{N} \frac{dt}{t} - \int_{1^-}^{N} \frac{d\{t\}}{t}$$
$$= \int_{1}^{N} \frac{dt}{t} - \int_{1}^{N} \frac{d\{t\}}{t} = \log N + 1 - \int_{1}^{N} \frac{\{t\}}{t^2}dt,$$

using that $\int_{1^-}^{N} = \int_1^N$. This confirms exercise 2.6 and provides a value for γ:

$$\lim_{N\to\infty}\left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{N} - \log N\right) \text{ exists and equals } \gamma := 1 - \int_1^\infty \frac{\{t\}}{t^2}dt. \tag{3.4}$$

This last integral converges since $\{t\} \in [0, 1)$ and so $\int_1^\infty \frac{\{t\}}{t^2}dt < \int_1^\infty \frac{1}{t^2}dt = 1$. The constant γ is known as the *Euler-Mascheroni constant.* Moreover

$$0 < \sum_{n=1}^{N} \frac{1}{n} - \log N - \gamma = \int_N^\infty \frac{\{t\}}{t^2}dt < \int_N^\infty \frac{dt}{t^2} = \frac{1}{N}.$$

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[2] More generally, without assumption about f, Abel wrote $a_n = S(n) - S(n-1)$, so that

$$\sum_{n=A+1}^{B} a_n f(n) = \sum_{n=A+1}^{B} f(n)(S(n) - S(n-1)),$$

which, with a little rearranging, equals $S(B)f(B) - S(A)f(A) - \sum_{n=A}^{B-1} S(n)(f(n+1) - f(n))$.

Exercise 3.1. Prove that $\sum_{n=1}^N \frac{1}{n} - \log N - \gamma$ is a decreasing function for $N \geq 1$.

Exercise 3.2. Prove that $\frac{1}{1} + \frac{1}{3} + \cdots + \frac{1}{2N-1} = \frac{1}{2}(\log N + \gamma) + \log 2 + O\left(\frac{1}{N}\right)$.

Exercise 3.3. Prove that $\prod_{n=1}^N (1 + \frac{1}{n})\, e^{-1/n} = e^{-\gamma}\left(1 + O\left(\frac{1}{N}\right)\right)$.

We can use the same method for

$$\log N! = \sum_{n=1}^N \log n = \int_{1^-}^N \log t\, d[t] = \int_1^N \log t\, dt - \int_1^N \log t\, d\{t\}$$
$$= N \log N - N + 1 + \int_1^N \frac{\{t\}}{t} dt$$

since $\int \log t\, dt = t(\log t - 1)$. The remaining integral is more challenging. The average of $\{t\}$ is $\frac{1}{2}$ so we would expect the integral to diverge, indeed to be roughly $\frac{1}{2}\log N$. We can see this by subtracting $\frac{1}{2}$ at each t so that

$$\int_1^N \frac{\{t\}}{t} dt = \frac{1}{2}\log N + \int_1^N \frac{\{t\} - \frac{1}{2}}{t} dt.$$

Now $\{t\} - \frac{1}{2}$ is 1-periodic and so is the integral $\int_0^T (\{t\} - \frac{1}{2}) dt$ since $\int_m^{m+1} (\{t\} - \frac{1}{2}) dt = \int_0^1 (t - \frac{1}{2}) dt = 0$. Therefore

$$\int_0^T (\{t\} - \tfrac{1}{2}) dt = \int_0^{\{T\}} (\{t\} - \tfrac{1}{2}) dt = \int_0^{\{T\}} (t - \tfrac{1}{2}) dt = \tfrac{1}{2}(\{T\}^2 - \{T\}).$$

Therefore we will evaluate the last integral by integrating by parts:

$$\int_1^N \frac{\{t\} - \frac{1}{2}}{t} dt = \int_1^N \frac{\{t\}^2 - \{t\}}{2t^2} dt,$$

which is convergent since the integrand is $\leq 1/4t^2$. Indeed

$$\int_1^\infty \frac{\{t\}^2 - \{t\}}{2t^2} dt = \tfrac{1}{2}\log(2\pi) - 1$$

though we will not prove this here. Combining these estimates and exponentiating yields *Stirling's formula*:

$$\boxed{N! = \left(\frac{N}{e}\right)^N \sqrt{2\pi N}\left(1 + O\left(\frac{1}{N}\right)\right).} \tag{3.5}$$

Exercise 3.4.

(a) Prove that $\binom{2N}{N} = \frac{4^N}{\sqrt{\pi N}}\left(1 + O\left(\frac{1}{N}\right)\right)$.

(b) Deduce that $\prod_{n=1}^N (1 + \frac{1}{2n})\, e^{-1/2n} = 2\sqrt{\frac{e^{-\gamma}}{\pi}}\left(1 + O\left(\frac{1}{N}\right)\right)$.

Exercise 3.5.* Prove that $\sum_{n \leq x} (\log(x/n))^k = \int_1^x (\log(x/t))^k dt + O((\log x)^k)$ and then that this equals $k!x + O((k + \log x)^k)$.

3.3. The sum of the reciprocals of the primes, revisited

We have proved that the sum of the reciprocals of the primes diverges. Now we apply partial summation to (2.14) to deduce a good estimate for the sum up to N:

Let[3] $S(t) = \sum_{p \le t} \frac{\log p}{p}$ with $M(t) = \log t$ so that

$$E(t) = S(t) - M(t) = O(1)$$

by (2.14). Therefore

$$\sum_{\substack{p \text{ prime} \\ p \le N}} \frac{1}{p} = \int_{2^-}^{N} \frac{dS(t)}{\log t} = \int_{2^-}^{N} \frac{dM(t)}{\log t} + \frac{E(t)}{\log t}\bigg|_{2^-}^{N} + \int_{2^-}^{N} \frac{E(t)}{t(\log t)^2} dt$$

$$= \int_{2^-}^{N} \frac{dt}{t \log t} + 1 + \frac{E(N)}{\log N} + \int_{2^-}^{N} \frac{E(t)}{t(\log t)^2} dt.$$

The last integral converges since

$$\int_{N}^{\infty} \frac{E(t)}{t(\log t)^2} dt \ll \int_{N}^{\infty} \frac{dt}{t(\log t)^2} = \frac{1}{\log N}$$

as $N \to \infty$. Therefore since $\frac{d}{dt} \log\log t = \frac{1}{t \log t}$ we deduce that

$$\boxed{\sum_{\substack{p \text{ prime} \\ p \le N}} \frac{1}{p} = \log\log N + c + O\left(\frac{1}{\log N}\right),} \tag{3.6}$$

where c is the constant $1 - \log\log 2 + \int_2^\infty \frac{E(t)}{t(\log t)^2} dt = .2614972\ldots$.[4]

Exercise 3.6. For c as in (3.6) prove that

$$\sum_{\substack{p \text{ prime}, e \ge 1 \\ p^e \le N}} \frac{1}{p^e} = \log\log N + c' + O\left(\frac{1}{\log N}\right) \quad \text{where } c' = c + \sum_p \frac{1}{p(p-1)}.$$

Exercise 3.7.*

(a) Prove *Mertens' Theorem* that there exists a constant γ such that

$$\prod_{p \le x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma}}{\log x} \left(1 + O\left(\frac{1}{\log x}\right)\right).$$

The constant γ equals the Euler-Mascheroni constant, though showing this is not part of the exercise. (Here we wrote "$p \le x$" as the subscript of the product, without adding "p prime", since that should be clear from the context.)

(b)† One can improve (2.13) by more sophisticated methods to $\sum_{n \le x} \frac{\Lambda(n)}{n} = \log x - \gamma + o(1)$. Deduce that $\sum_{p \le x} \frac{\log p}{p-1} = \log x - \gamma + o(1)$.

Exercise 3.8.* Assume that $\sum_{n \le x} \frac{\Lambda(n)}{n} = \log x + c + o(1)$ for some constant c. Using partial summation, deduce the Prime Number Theorem in the form $\psi(x) \sim x$.

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[3] Taking $a_n = \frac{\log p}{p}$ if $n = p$ is prime, and $a_n = 0$ otherwise, with $f(t) = \frac{1}{\log t}$.

[4] $\log\log x$ goes to infinity as $x \to \infty$ but slowly. To give you some idea of how slow, there are about 10^{80} atoms in the observable universe and $\log\log(10^{80}) \approx 5.216$.

3.4. The error term in the Prime Number Theorem

We now show that the equivalence in Lemma 2.1 can be extended to a wide range of plausible error terms:

PROPOSITION 3.1 (Error terms in the Prime Number Theorem). *If* $\pi(x) = \mathrm{Li}(x) + O(E_\pi(x))$ *where* $E_\pi(x)$ *is increasing, then* $\Theta(x) = x + O(E_\pi(x)\log x)$. *If* $\Theta(x) = x + O(E_\Theta(x))$ *where* $E_\Theta(x)$ *is increasing and* $E_\Theta(x) \gg \sqrt{x}$, *then*

$$\pi(x) = \mathrm{Li}(x) + O\left(\frac{E_\Theta(x)}{\log x}\right).$$

Thus $E_\Theta(x) \asymp E_\pi(x)\log x$ under the various hypotheses of the proposition.

PROOF. If $\pi(t) - \mathrm{Li}(t) \ll E_\pi(t)$, then

$$\begin{aligned}\Theta(x) - x &= \int_{2^-}^{x} \log t \, d(\pi(t) - \mathrm{Li}(t)) - 2 \\ &= (\pi(x) - \mathrm{Li}(x))\log x - \int_2^x \frac{\pi(t) - \mathrm{Li}(t)}{t} dt + O(1) \\ &\ll 1 + E_\pi(x)\left(\log x + \int_2^x \frac{1}{t} dt\right) \ll E_\pi(x)\log x\end{aligned}$$

as $E_\pi(t)$ is increasing. In the other direction, if $\Theta(t) - t \ll E_\Theta(t)$, then

$$\begin{aligned}\pi(x) - \mathrm{Li}(x) &= \int_{\sqrt{x}}^{x} \frac{d(\Theta(t) - t)}{\log t} + O\left(\frac{\sqrt{x}}{\log x}\right) \\ &= \frac{\Theta(x) - x}{\log x} + \int_{\sqrt{x}}^{x} \frac{\Theta(t) - t}{t(\log t)^2} dt + O\left(\frac{\sqrt{x}}{\log x}\right) \\ &\leq E_\Theta(x)\left(\frac{1}{\log x} + \int_{\sqrt{x}}^{x} \frac{1}{t(\log t)^2} dt\right) \ll \frac{E_\Theta(x)}{\log x}\end{aligned}$$

since $E_\Theta(t)$ is increasing. □

Exercise 3.9.* Prove that $\pi(x) = \mathrm{Li}(x) + O(\sqrt{x}\log x)$ if and only if $\psi(x) = x + O(\sqrt{x}(\log x)^2)$.

Exercise 3.10.* Using Chebyshëv's bounds (2.10) and partial summation, prove that

$$\sum_{p>x} \frac{1}{p^{1+\frac{1}{\log x}}} = O(1).$$

This estimate will be useful later.

3.5. The value of a limit

The Prime Number Theorem states that $\psi(x) \sim x$, but we have not proved that yet. Here we will show that if $\psi(x)/x$ tends to a limit as $x \to \infty$, then the Prime Number Theorem follows: Suppose that $\lim_{x\to\infty} \psi(x)/x$ exists and equals L. The Chebyshëv estimates (2.10) show that $\log 2 \leq L \leq \log 4$, but they yield no better estimate for L. Our assumption implies $\Theta(x) = \{L + o(1)\}x$. Using partial summation we obtain

$$\sum_{p \le x} \frac{\log p}{p} = \int_{2^-}^{x} \frac{d\Theta(t)}{t} = L \int_{2^-}^{x} \frac{dt}{t} + \frac{E(t)}{t}\bigg|_{2^-}^{x} + \int_{2^-}^{x} \frac{E(t)}{t^2} dt$$
$$= L \log x + O(1) + o(\log x),$$

where $E(t) = \Theta(t) - Lt = o(t)$. Comparing this with (2.14), we deduce that $L = 1$ and therefore the Prime Number Theorem holds.

Since we do not know that $\psi(x)/x$ tends to a limit we cannot deduce the Prime Number Theorem. However, we have shown that if $\psi(x)/x$ does tend to a limit, then that limit must be 1. Hence the "difficulty" in proving the Prime Number Theorem rests in showing that $\lim_{x\to\infty} \psi(x)/x$ exists.

CHAPTER 4

What should be true about primes?

Before proceeding to prove estimates about the distribution of primes we should get a better sense of what we believe is true. In this chapter we introduce a heuristic technique which, when applied suitably, yields a wide range of predictions that motivate some of the main goals of analytic number theory.

We begin with Gauss's suggestion that the density of primes *near* x is around $1/\log x$, which allowed us to predict the Prime Number Theorem (see section 1.1). How else can Gauss's suggestion be used to make predictions? For example, in an interval of length y around x, we might expect that

$$\#\{p \text{ prime} : x < p \le x + y\} \text{ is about } \frac{y}{\log x}. \tag{4.1}$$

However, this claim is difficult to interpret if $y = \frac{1}{2}\log x$ as one can't have half a prime. So, what do we mean? Can we be any more precise?

4.1. The Gauss-Cramér model for the primes

In 1936, the great probabilist Harald Cramér interpreted Gauss's suggestion in creating the following probabilistic model to study the sequence of primes: Let $X_3, X_4, \dots$ be an infinite sequence of independent random variables for which[1]

$$\text{Prob}(X_n = 1) \;=\; \frac{1}{\log n} \quad \text{and} \quad \text{Prob}(X_n = 0) \;=\; 1 - \frac{1}{\log n}.$$

This defines a probability space on all infinite sequences of 0's and 1's. The indicator function for the odd primes is such a sequence:

	1	0	1	0	1	0	0	0	1	0	1	0	0	0	1	0...
corresponding to	3		5		7				11		13				17	

Cramér proposed that one should think of the sequence of 0's and 1's corresponding to the primes as a "typical" sequence of 0's and 1's, under this probability measure, so that anything that can be said with probability 1 for the space of such sequences should be true for the sequence of primes.

For example, let $S_N := X_3 + \cdots + X_N$ which has mean[2]

$$\mu_N := \mathbb{E}(S_N) = \mathbb{E}\left(\sum_{n=3}^{N} X_n\right) = \sum_{n=3}^{N} \mathbb{E}(X_n) = \sum_{n=3}^{N} \frac{1}{\log n} = \text{Li}(N) + O(1)$$

by exercise 1.11. (Here "$\mathbb{E}(\cdot)$" stands for "Expectation".) To employ Cramér's proposal we need to determine whether S_N is usually "close" to its expected value,

[1] We cannot include X_1 and X_2 since $\frac{1}{\log n} > 1$ for $n = 1$ and 2.

[2] A random variable S has mean value $\mu := \mathbb{E}(S)$ with *variance* $\sigma^2 := \mathbb{E}((S - \mu)^2)$.

μ_N. To do this we study the *variance*, a key quantity in probability theory:[3]

$$\begin{aligned}\mathbb{E}(|S_N-\mu_N|^2) &= \mathbb{E}\left(\left(\sum_{n=3}^{N} X_n - \sum_{n=3}^{N}\frac{1}{\log n}\right)^2\right) = \sum_{n=3}^{N}\mathbb{E}\left(\left(X_n-\frac{1}{\log n}\right)^2\right)\\ &= \sum_{n=3}^{N}\left(\mathbb{E}\left(X_n^2\right)-\frac{1}{(\log n)^2}\right) = \sum_{n=3}^{N}\left(\frac{1}{\log n}-\frac{1}{(\log n)^2}\right)\\ &= \frac{N}{\log N}+O(1)\end{aligned}$$

by exercise 1.14.

Let $H(t) := \mathbb{P}(S_N-\mu_N\le t)$ (where $\mathbb{P} = $ Prob) so that by partial summation

$$\mathbb{E}(|S_N-\mu_N|^2)=\int_{t=-\infty}^{\infty} t^2 dH(t) \ge \left(\int_{t=-\infty}^{-T}+\int_{t=T}^{\infty}\right) T^2 dH(t) = T^2\mathbb{P}(|S_N-\mu_N|>T)$$

for any $T>0$. Therefore, for any fixed $\epsilon>0$ we have

$$\mathbb{P}(|S_N-\mu_N|>N^{\frac12})\cdot N \le \mathbb{E}(|S_N-\mu_N|^2) \ll \frac{N}{\log N},$$

so that $\mathbb{P}(|S_N-\mu_N|>N^{\frac12}) \ll \frac{1}{\log N}$; that is,

$$|S_N-\mu_N|\le N^{\frac12} \text{ with probability } 1-O\left(\frac{1}{\log N}\right)=1-o_N(1). \tag{4.2}$$

This is the continuous version of Chebyshëv's inequality (see appendix A). Thus Cramér's proposal suggests that

$$|\pi(N)-\mathrm{Li}(N)|\le N^{\frac12}.$$

This fits very well with what we observed about the data for the sequence of primes. Moreover we will see (in (22.5) and Theorem 22.1) that this bound is more or less equivalent to the famous "Riemann Hypothesis" (Conjecture 18.1).

Exercise 4.1. Show, more precisely, that for any fixed $\tau>1$ we have

$$\mathbb{P}\left(|S_N-\mu_N|\ge \tau\sqrt{\tfrac{N}{\log N}}\right) \ll \tfrac{1}{\tau^2},$$

so that this probability $\to 0$ as $\tau\to\infty$.

We can make more precise guesses about $\pi(x)$ using deeper tools of probability theory. The *Central Limit Theorem* tells us that a sum of independent random variables, satisfying certain reasonable hypotheses, should converge to the normal distribution (see the discussion in section A.3). In particular Lyapunov's Central Limit Theorem implies that for any fixed $T\in\mathbb{R}$ we have

$$\mathbb{P}\left(S_N-\mu_N\ge T\sqrt{\tfrac{N}{\log N}}\right)=\{1+o_N(1)\}\frac{1}{\sqrt{2\pi}}.\int_T^{\infty} e^{-u^2/2}du. \tag{4.3}$$

Cramér's proposal therefore suggests that the error term $\pi(N)-\mathrm{Li}(N)$ is distributed like a normal distribution with mean 0 and variance $\frac{N}{\log N}$ as N varies.[4]

[3] See appendix A to explain some of the steps.

[4] A more sophisticated heuristic (discussed in section 34.2 based on a deep understanding of the zeros of $\zeta(s)$) suggests that this is untrue.

Kolmogorov's *law of the iterated logarithm* tells us how large $|S_N - \mu_N|$ can get (as a function of N). Via Cramér's proposal, this suggests that

$$|\pi(N) - \mathrm{Li}(N)| \lesssim \left(\frac{2N \log\log N}{\log N}\right)^{1/2} \tag{4.4}$$

and that there is no significantly smaller upper bound.[5]

These arguments can be modified to primes in short intervals: Cramér's proposal suggests that if $y = X^c$ for some $c \in (0,1)$, then the error term in

$$\#\{\text{Primes } p \in (x, x+y]\} - \int_x^{x+y} \frac{dt}{\log t},$$

as x varies between X and $2X$, should be distributed like a normal distribution with mean 0 and variance $\sim \frac{y}{\log x}$.[6,7]

A consequence of this is that for any fixed $\epsilon > 0$ we have, with probability $\to 1$ as $x \to \infty$,

$$\left| \sum_{n=x+1}^{x+y} X_n - \int_x^{x+y} \frac{dt}{\log t} \right| \le y^{\frac{1}{2}}.$$

Cramér's proposal replaces $\sum_{n=x+1}^{x+y} X_n$ by $\pi(x+y) - \pi(x)$. If $y \le x$, then $\log t \sim \log x$ for all $t \in [x, x+y]$ and so $\int_x^{x+y} \frac{dt}{\log t} \sim \frac{y}{\log x}$. This is larger than the bound $y^{\frac{1}{2}}$ for $y \ge (\log x)^{2+\epsilon}$ if $\epsilon > 0$ is sufficiently small. Therefore something like (4.1) should hold provided $y \ge (\log x)^{2+\epsilon}$, a rather short interval. However, we are far from proving anything like this for the primes themselves. Such an analysis can be developed to even suggest that

$$\pi(x+y) - \pi(x) \sim \frac{y}{\log x} \tag{4.5}$$

for 100% of the intervals $(x, x+y]$ with $x \le X$ where $y = y(x)$ is a function of x for which $y/\log x \to \infty$ as $x \to \infty$. The "100%" here does not mean the same thing as "all". This is 100% in the sense that this estimate is true for a certain proportion of intervals up to X, and this proportion tends to 100% as $X \to \infty$. Thus we say that (4.5) holds for "*almost all*" x for such values of y (rather than "all"). Selberg proved this is true when $y/(\log x)^2 \to \infty$ as $x \to \infty$, assuming the Riemann Hypothesis (see [**A-6**, section 30.2]).

[5] A more sophisticated heuristic (discussed in section 34.4, again based on understanding the zeros of $\zeta(s)$) suggests that the upper bound should be significantly smaller, the $(\log N)^{-\frac{1}{2}+o(1)}$ replaced by $(\log N)^{-1+o(1)}$. Nonetheless Cramér's heuristic provides a guess which is close to what we believe.

[6] We choose to vary x in the interval $(X, 2X]$, rather than in $[1, X]$, so that $y(x)$ is close to $y(X)$ for our choice of $y(\cdot)$ and so does not vary much for these x-values. This makes it then straightforward to generalize such predictions to all x-values.

[7] A more sophisticated heuristic (discussed in section 34.6 based on understanding the zeros of $\zeta(s)$) suggests that this satisfies a normal distribution with the slightly smaller variance $\sim \frac{y}{\log x}(1 - \frac{\log y}{\log x})$.

4.2. Short intervals

Cramér's heuristic suggests that intervals of length $\frac{1}{2}\log x$ contain half a prime on average. More meaningfully it suggests that the number of primes in intervals length $\frac{1}{2}\log x$ follow a Poisson distribution with parameter $\lambda = \frac{1}{2}$.[8] This means that for any fixed real number $\lambda > 0$ and any fixed integer $k \geq 0$,

$$\mathbb{P}((x, x+\lambda\log x] \text{ contains exactly } k \text{ primes, where } x \leq X) \to \frac{e^{-\lambda}\lambda^k}{k!}$$

as $X \to \infty$. This probability is maximized at the integers k which are closest to λ, the average. Indeed, for a Poisson distribution with (large) parameter λ, one can show that $k = \lambda + O(\lambda^{1/2+o(1)})$ with probability $1 + o_{\lambda\to\infty}(1)$ (see exercise A.3(b)). In particular exercise A.3(a) confirms (as discussed at the end of the last section) that (4.5) holds for "almost all" x, provided $y/\log x \to \infty$ and $x \to \infty$.

4.3. Largest gaps between primes

One can also turn to the Gauss-Cramér model to guess how big y needs to be, in terms of x, to *guarantee* a prime in $(x, x+y]$ for every x. That is tantamount to asking for the largest gaps between consecutive primes. Now, if x and y are integers, then, since the X_n are independent,

$$\mathbb{P}(X_n = 0 \text{ for all } n \in (x, x+y]) = \prod_{n=x+1}^{x+y} \mathbb{P}(X_n = 0) = \prod_{n=x+1}^{x+y} \left(1 - \frac{1}{\log n}\right).$$

If y is not large compared to x, then $\log n$ is close to $\log x$,[9] and we can approximate this product by

$$\left(1 - \frac{1}{\log x}\right)^y \quad \text{which is roughly } e^{-y/\log x}.$$

Therefore if $X < x \leq 2X$ and y is "small", then[10]

$$\mathbb{P}(X_n = 1 \text{ for some } n \in (x, x+y]) \approx 1 - e^{-y/\log X}.$$

We want to select y so that this is very close to 1; certainly y must be substantially larger than $\log X$. There are X intervals of the form $(x, x+y]$ with $X < x \leq 2X$, and so the probability that all of them contain at least one prime is therefore roughly[11]

$$\left(1 - e^{-y/\log X}\right)^X \approx e^{-Xe^{-y/\log X}}.$$

This quantity is very close to 1 if y is a little bit bigger than $(\log X)^2$, and it is very close to 0 if y is a little bit smaller than $(\log X)^2$. Cramér developed this into a formal proof that, with probability $1 + o_{X\to\infty}(1)$, if $n_1 < n_2 < n_3 < \cdots$ are the indices n for which $X_n = 1$, then

$$\max_{k:\ n_k \leq X} (n_{k+1} - n_k) \sim (\log X)^2.$$

[8] See appendix A.2 for details.

[9] More precisely, $\log n = \log(x + O(y)) = \log x + O(y/x)$.

[10] The symbol "$\approx$" means "is approximately equal to" or "is roughly", but it has no precise technical meaning, but rather it is used to help get an understanding of what to believe. We vary x in the interval $(X, 2X]$ here so that the value of $\log x$ varies by no more than $O(1)$.

[11] To be more precise with the probability theory, we should really work with X/y disjoint intervals, though this will make an unimportant difference to the outcome of this calculation.

Therefore Cramér conjectured that the largest gap between consecutive primes $p_n < p_{n+1} \le X$ should be about $(\log p_n)^2$. In other words, if $p_1 = 2 < p_2 = 3 < \cdots$ is the sequence of prime numbers, then

$$\max_{p_n \le x} (p_{n+1} - p_n) \sim (\log x)^2.$$

Here are the record-breaking gaps, in terms of the ratio of these two quantities:

p_n	$p_{n+1} - p_n$	$(p_{n+1} - p_n)/\log^2 p_n$
113	14	.6264
1327	34	.6576
31397	72	.6715
370261	112	.6812
2010733	148	.7026
20831323	210	.7395
25056082087	456	.7953
2614941710599	652	.7975
19581334192423	766	.8178
218209405436543	906	.8311
1693182318746371	1132	.9206

TABLE 4.1. (Known) record-breaking gaps between primes.

Evidently the constant is creeping upwards, but will it reach 1?[12] We don't know.

4.4. Even primes

There is only one even prime yet the Gauss-Cramér model suggests that the number of even primes up to x is $\sim \sum_{m=1}^{x/2} \frac{1}{\log 2m} \sim \frac{x}{2\log x}$.

The Gauss-Cramér model also suggests that the number of prime pairs $p, p+k$ with $p \le N$ is about

$$\mathbb{E}\left(\sum_{n=3}^{N} X_n X_{n+k}\right) = \sum_{n=3}^{N} \mathbb{E}\left(X_n\right)\mathbb{E}\left(X_{n+k}\right) = \sum_{n=3}^{N} \frac{1}{(\log n)\log(n+k)}$$

which is $\sim \frac{N}{(\log N)^2}$. One gets the same number whether $k = 1$ (when there is just one such prime pair, namely 2 and 3) or $k = 2$ (when we expect infinitely many), so the model gives an incorrect prediction, evidently because it again does not take account of divisibility by 2.

In the next chapter we will modify the Gauss-Cramér model to properly take account of divisibility by 2, as well as divisibility by other small primes.

[12] And will it go beyond? This is suggested by the modified heuristic given in chapter 5 — see the end of section 5.2.

CHAPTER 20

The use of Perron's formula

In this chapter we work through several examples in analytic number theory using Perron's formula, some of which have been worked on earlier by different methods.

20.1. An arithmetic function as a sum of residues

Example 20.1. Counting the number of integers $\leq x$:

$$\sum_{n\leq x}^{\flat} 1 = \frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \zeta(s)\frac{x^s}{s}ds.$$

This integrand has a simple pole at $s = 1$ because $\zeta(s)$ does. To calculate the residue we need to determine $\lim_{s\to 1}(s-1)\zeta(s)\frac{x^s}{s}$. Here we know that $\lim_{s\to 1}(s-1)\zeta(s) = 1$, and so our residue is $x^1/1 = x$. The pole at $s = 0$ has residue $\zeta(0) = -\frac{1}{2}$ (see exercise 18.4). Therefore the sum of the residues is $x - \frac{1}{2}$, which is exactly equal to our original sum when $x \in \mathbb{Z}$, and is always close: Indeed, if $x \notin \mathbb{Z}$, then the original sum is $[x]$ and so the difference is $x - [x] - \frac{1}{2} = \{x\} - \frac{1}{2}$, which has a Fourier series that we investigated in section 16.1.

Exercise 20.1. Approximate $\sum_{n\leq x}^{\flat} \log n$ by a sum of residues. Compare this to Stirling's formula, (3.5), using that $-\zeta'(0) = \frac{1}{2}\log 2\pi$.

Example 20.2. In section 6.3 we introduced Dirichlet's hyperbola trick in estimating

$$\sum_{n\leq x}^{\flat} \tau(n) = \frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \zeta(s)^2\frac{x^s}{s}ds.$$

This has a pole at $s = 1$ of order 2. Calculating the residues at a kth-order pole s_0 of a function $f(s)$, when $k > 1$, is more complicated than simply calculating $\lim_{s\to s_0}(s - s_0)f(s)$, as we did in the previous example. Now we have to calculate the coefficient of the $1/(s-1)$ term in the Laurent series for $f(s)$ which is the same as the coefficient of the $(s - 1)^{k-1}$ term in $(s - 1)^k f(s)$. If we let $s \to 1$, then one approach is to calculate the Taylor series of $(s - 1)^k f(s)$ within an error of $O_{s\to 1}((s - 1)^k)$. In our example $k = 2$ and we have

$$((s-1)\zeta(s))^2 = (1 + \gamma(s-1) + O((s-1)^2))^2 = 1 + 2\gamma(s-1) + O((s-1)^2).$$

For the other terms one has $\frac{1}{s} = \frac{1}{1+(s-1)} = 1 - (s - 1) + O((s - 1)^2)$ and

$$x^s = x\cdot x^{s-1} = xe^{(s-1)\log x} = x(1 + (s-1)\log x + O((s-1)^2)).$$

Combining these calculations we deduce that

$$((s-1)\zeta(s))^2\frac{x^s}{s} = x(1+(\log x+2\gamma-1)(s-1)+O((s-1)^2));$$

and so the residue of $\zeta(s)^2\frac{x^s}{s}$ at $s=1$ is $x(\log x+2\gamma-1)$. This is precisely the main term we found in section 6.3 by elementary methods.

Exercise 20.2. A simple generalization is given by the mean value of $d_k(n)$ for k an integer >2. Apply Perron's formula, and show that the residue at $s=1$ of the corresponding integral yields a main term of the form $xP_k(\log x)$ where $P_k(t)$ is a polynomial of degree $k-1$ whose coefficients can be determined in terms of the coefficients of the Laurent series for $\zeta(s)$. (This was proved by a complicated elementary method in exercise 6.11.)

Exercise 20.3. Suppose that $A(s)$ is analytic at $s=1$. Prove that the residue of $A(s)\zeta(s)^k x^s$ at $s=1$ is a polynomial in $\log x$ of degree $k-1$, whose coefficients can be given explicitly in terms of the coefficients of the Taylor series of $A(s)$ and of $((s-1)\zeta(s))^k$.

EXAMPLE 20.3. If $\alpha\in\mathbb{C}$ and $\zeta(s)^\alpha=\sum_{n\geq 1} d_\alpha(n)/n^s$ for $\mathrm{Re}(s)>1$ (as defined in section 6.1), then we evidently have

$$\sideset{}{^\flat}\sum_{n\leq x} d_\alpha(n) = \frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \zeta(s)^\alpha\frac{x^s}{s}ds.$$

Now $\zeta(s)^\alpha$ looks like $(s-1)^{-\alpha}$ near $s=1$, and so it cannot be meromorphic if $\mathrm{Re}(\alpha)$ is not an integer. This is therefore a much more challenging question that we will return to in [**A-6**, section 34.1].

EXAMPLE 20.4. In section 14.2 we counted the squarefree and powerful numbers $\leq x$. The Euler products of the corresponding Dirichlet series have pth terms

$$1+\frac{1}{p^s} \quad\text{and}\quad 1+\frac{1}{p^{2s}}+\frac{1}{p^{3s}}+\cdots = 1+\frac{1}{p^s(p^s-1)},$$

respectively, and these can be rewritten as

$$\frac{1-p^{-2s}}{1-p^{-s}} \quad\text{and}\quad \frac{1-p^{-6s}}{(1-p^{-2s})(1-p^{-3s})}$$

and so have the Dirchlet series

$$\zeta(s)/\zeta(2s) \quad\text{and}\quad \zeta(2s)\zeta(3s)/\zeta(6s).$$

Therefore the number of squarefree integers $\leq x$ is

$$\sideset{}{^\flat}\sum_{n\leq x} \mu^2(n) = \frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \frac{\zeta(s)}{\zeta(2s)}\frac{x^s}{s}ds,$$

which has a pole with residue $x/\zeta(2)$ at $s=1$ and poles at the zeros of $\zeta(2s)$ that are not also a zero of $\zeta(s)$. The Riemann Hypothesis (Conjecture 18.1) claims that the zeros of $\zeta(2s)$ all lie on the line $\mathrm{Re}(s)=\frac{1}{4}$ so the number of squarefree integers $\leq x$ should be $x/\zeta(2)$ plus a complicated term of size around $x^{1/4}$.

For the number of powerful numbers up to x we have

$$\sideset{}{^\flat}\sum_{\substack{n\leq x\\ n \text{ powerful}}} 1 = \frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \frac{\zeta(2s)\zeta(3s)}{\zeta(6s)}\frac{x^s}{s}ds.$$

In this case the first pole that we encounter is when $2s = 1$, that is, at $s = \frac{1}{2}$. In this case the residue is $\lim_{s\to\frac{1}{2}}(2s-1)\zeta(2s)\zeta(3s)x^s/2s\zeta(6s) = c_1x^{1/2}$ where $c_1 = \zeta(3/2)/\zeta(3)$, the main term we obtained in section 14.2. The next pole that we encounter is when $3s = 1$, with residue $c_2x^{1/3}$ where $c_2 = \zeta(2/3)/\zeta(2)$. Then the next poles occur at the zeros of $\zeta(6s)$ so we might hope for an error term of size around $x^{1/12}$. Therefore we might guess that the number of powerful numbers up to x is $c_1x^{1/2} + c_2x^{1/3} + O(x^{1/12+o(1)})$.

20.2. The integrals for counting primes

One can determine $\pi(x)$ via Perron's formula using the identity

$$\sum_p \frac{1}{p^s} = \sum_{m\geq 1} \frac{\mu(m)}{m} \log\zeta(ms),$$

leading to a sum of infinitely many integrals. The $m = 1$ term leads to the integral

$$\frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \log\zeta(s)\frac{x^s}{s}ds.$$

The pole of $\zeta(s)$ at $s = 1$ means we will need to understand the behavior of $\log(s-1)$ around $s = 1$, which is doable,[1] but it is easier to avoid such logarithmic singularities (and branch cuts, etc.) by replacing 1 at each prime by von Mangoldt's function $\Lambda(n)$. Therefore we prefer to work with

$$\psi^\flat(x) = -\frac{1}{2\pi i}\int_{s:\ \mathrm{Re}(s)=2} \frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds \tag{20.1}$$

(as in (19.3)). Since $\zeta(s)$ is a meromorphic function (in particular, it has a Laurent series at each point), $\zeta'(s)/\zeta(s)$ is also a meromorphic function. Moreover, since $\zeta(s)$ has a simple pole at $s = 1$, the function $\frac{-\zeta'(s)}{\zeta(s)}$ also has a simple pole at $s = 1$ with residue 1. Therefore the residue at $s = 1$ is $1 \cdot x^1/1 = x$. If we move the contour back to the left, then we have to determine all the poles of

$$\frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s},$$

which, besides the poles at $s = 0$ and $s = 1$, are precisely the zeros of $\zeta(s)$. The Riemann Hypothesis states that all such zeros ρ satisfy $\mathrm{Re}(\rho) \leq \frac{1}{2}$, and there are some ρ with $\mathrm{Re}(\rho) = \frac{1}{2}$, and so since $|x^\rho| = x^{\mathrm{Re}(\rho)}$, we expect a complicated error term of size around $x^{1/2}$.

[1] Indeed Riemann took this more difficult approach and made it work, but hindsight is 20/20.

CHAPTER 21

Riemann's plan for proving the Prime Number Theorem

In 1859, the great geometer Riemann took up the challenge in [**67**] of finding a method to estimate $\pi(x) = \#\{\text{primes} \le x\}$. Whereas previous approaches were relatively direct, based upon elementary number theory and combinatorial principles, or the theory of quadratic forms, Riemann described a surprising approach to the problem using complex analysis. His approach makes predictions that are even better than Gauss's prediction, since it suggests further terms to compensate for the overcount we saw in the data in the table in Figure 1.1. Moreover it led to an outline of a proof of the Prime Number Theorem, which was eventually completed by others.

Although his approach seemed far away from the realm in which the original problem lived, it was eventually realized that it implies that number theory and complex analysis are inseparable, inspiring the subject we now call *analytic number theory*.

21.1. A method to accurately estimate the number of primes

Beginning with Riemann's prediction that

$$\psi(x) = \sum_{k\ge 1} \Theta(x^{1/k}) \text{ is very well approximated by } x$$

(a prediction we will justify below), we then deduce by partial summation that

$$\sum_{k\ge 1} \frac{1}{k}\pi(x^{1/k}) \text{ is very well approximated by } \mathrm{Li}(x).$$

By Möbius inversion (as in exercise 2.5) we then deduce that

$$\pi(x) \text{ is very well approximated by } \sum_{m\ge 1} \frac{\mu(m)}{m}\,\mathrm{Li}(x^{1/m}).$$

The sum of the terms with $m > 2$ will be much smaller than $x^{1/2-\epsilon}$, so Riemann's methods yields the prediction that

$$\pi(x) \approx \mathrm{Li}(x) - \frac{1}{2}\,\mathrm{Li}(x^{1/2}),$$

revealing a secondary term not found in Gauss's prediction. Reviewing our data (where the column headed "Riemann" refers to $\mathrm{Li}(x) - \frac{1}{2}\,\mathrm{Li}(\sqrt{x}) - \pi(x)$, while the column headed "Gauss" refers to $\mathrm{Li}(x) - \pi(x)$ as before) we have Table 21.1.

x	#{primes $\leq x$}	Gauss	Riemann
10^8	5761455	753	131
10^9	50847534	1700	-15
10^{10}	455052511	3103	-1711
10^{11}	4118054813	11587	-2097
10^{12}	37607912018	38262	-1050
10^{13}	346065536839	108970	-4944
10^{14}	3204941750802	314889	-17569
10^{15}	29844570422669	1052618	76456
10^{16}	279238341033925	3214631	333527
10^{17}	2623557157654233	7956588	-585236
10^{18}	24739954287740860	21949554	-3475062
10^{19}	234057667276344607	99877774	23937697
10^{20}	2220819602560918840	222744643	-4783163
10^{21}	21127269486018731928	597394253	-86210244
10^{22}	201467286689315906290	1932355207	-126677992
10^{23}	1925320391606803968923	7236148412	1034233576
10^{24}	18435599767349200867866	17146907278	-1657067862
10^{25}	176846309399143769411680	55160980939	-1830820885
10^{26}	1699246750872437141327603	155891678121	-17141144784
10^{27}	16352460426841680446427399	508666658006	-17518534194
10^{28}	157589269275973410412739598	1427745660374	-174725372473
10^{29}	1520698109714272166094258063	4551193622464	-335739334094

TABLE 21.1. Primes up to various x, and Gauss's and Riemann's predictions.

Riemann's prediction fares better than Gauss's prediction. The error term in Riemann's prediction takes both positive and negative values which suggests that any further terms will need to change sign, back and forth, as x grows.

21.2. Linking number theory and complex analysis

In section 2.4 we discussed the key link between primes and the Riemann zeta-function: We have

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 1} \frac{\Lambda(n)}{n^s} \quad \text{for } \mathrm{Re}(s) > 1$$

and so $\psi(x) := \sum_{n\leq x} \Lambda(n)$ is the sum of the coefficients of $-\zeta'(s)/\zeta(s)$ over all $n \leq x$. One can extract the sum of the coefficients of a Dirichlet series (under the right conditions) by using Perron's formula, and obtain, as in (20.1),

$$\psi^\flat(x) = -\frac{1}{2\pi i}\int_{s:\mathrm{Re}(s)=2} \frac{\zeta'(s)}{\zeta(s)}\, \frac{x^s}{s}\, \mathrm{d}s, \tag{21.1}$$

where $\psi^\flat(x) = \psi(x) - \frac{1}{2}\Lambda(x)$ (where $\Lambda(x) := 0$ when x is not an integer).

To evaluate this integral we want to pull the contour of integration far to the left (as the integrand gets smaller to the left), using the ideas of complex analysis. We can do so because the integrand is a function that makes sense on the whole complex plane following the discussion of section 18.4. If we can select contours which make a negligible contribution, then, by Cauchy's Theorem, the value of the integral will be given by the sum of the residues of the integrand to the left of $\mathrm{Re}(s) = 2$ plus a small "error term".

The poles of the integrand $-\frac{\zeta'(s)}{\zeta(s)}\ \frac{x^s}{s}$ are at $s = 0$ (the only pole of $\frac{x^s}{s}$), $s = 1$ (the only pole of $\zeta'(s)$), and the zeros of $\zeta(s)$. If $\zeta(s)$ has a zero of order $m_\rho \geq 1$ at $s = \rho$, then $\frac{\zeta'(s)}{\zeta(s)}$ has a simple pole at $s = \rho$ with residue m_ρ (as shown in section 18.4). Therefore:

- When $s = 1$ (so that $m_1 = -1$) we obtain the residue x.
- When $s = 0$ we obtain the residue $-\frac{\zeta'(0)}{\zeta(0)}$, a constant which equals $-\log 2\pi$.
- At each zero $s = \rho$ of $\zeta(s)$, we obtain the residue $-m_\rho\,\frac{x^\rho}{\rho}$.

Therefore if we can shift contours appropriately all the way back as $\mathrm{Re}(s) \to -\infty$, then we can deduce the following:

THEOREM 21.1 (The explicit formula). *For any $x \geq 2$ we have*

$$\psi^\flat(x) = \sideset{}{^\flat}\sum_{\substack{p \text{ prime},\ k\geq 1 \\ p^k \leq x}} \log p = x - \sum_{\rho:\ \zeta(\rho)=0} \frac{x^\rho}{\rho} - \log 2\pi. \tag{21.2}$$

Here if ρ is a zero of $\zeta(s)$ of order m_ρ, then there are m_ρ terms for ρ in the sum.

This is a rather surprising formula: a discrete function $\psi^\flat(x)$ being given as a sum of continuous functions. The right-hand side looks quite a bit like a Fourier series when one writes $\rho = \beta + i\gamma$ to obtain

$$x^\rho = x^\beta \cdot e^{i\gamma \log x} = x^\beta(\cos(\gamma \log x) + i \sin(\gamma \log x)).$$

This is how Riemann viewed this formula and it is a useful perspective when trying to understand the development of the error term in the Prime Number Theorem.

The left-hand side jumps in value each time we encounter a prime power (as x increases) whereas the right-hand side is a sum of continuous functions. To have equality the right-hand side must therefore be an infinite sum; that is, there must be infinitely many zeros of $\zeta(s)$, and the sum cannot be absolutely convergent. Therefore we must be concerned about the order in which we take the terms in our infinite sum: It should be added up in the order of ascending values of $|\rho|$.

It is difficult to believe that there can be a formula like (21.2), an exact expression for the number of primes up to x in terms of the zeros of an analytic continuation. You can see why Riemann's work stretched people's imagination and had such a profound impact. Riemann's formula (21.2) is a little hard to appreciate at first glance, so we will be more explicit in section 21.3 after developing and explaining one or two key details.

The trivial zeros of $\zeta(s)$ lie at $-2, -4, \ldots$ and so their contribution to (21.2) is

$$-\sum_{n\geq 1} \frac{x^{-2n}}{-2n} = -\frac{1}{2}\log(1 - x^{-2}).$$

Therefore we can rewrite the explicit formula, (21.2), as

$$\psi^\flat(x) = x - \sum_{\substack{\rho:\ \zeta(\rho)=0 \\ 0\leq \mathrm{Re}(\rho)\leq 1}} \frac{x^\rho}{\rho} - \frac{1}{2}\log(1 - x^{-2}) - \log 2\pi. \tag{21.3}$$

To pull the contour of integration in the integral in (21.1) to the left we begin by truncating at some height T so the integral now runs from $2-iT$ to $2+iT$, and we need to bound the contribution of the integral on the pruned contours. We next create a closed rectangular box as in Figure 21.1:

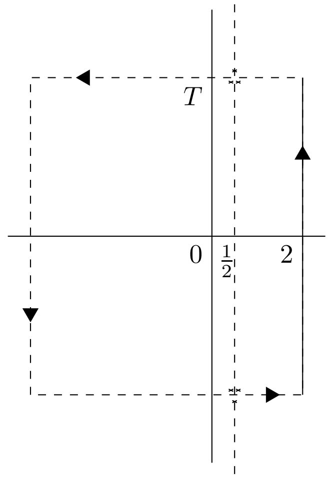

FIGURE 21.1. Pulling the contour to the left and looking for poles.

The horizontal lines might pass through, or too near to, a zero of $\zeta(s)$ and then the integrand in (21.1) will pass close to a pole of $\zeta'(s)/\zeta(s)$, so the contribution along the contour might become large. Therefore we wish to thread our horizontal contours, at the top and at the bottom, so they keep well away from any zeros of $\zeta(s)$. How far away will depend on how dense the zeros are, which can be understood through (18.8).

Exercise 21.1. Explain why there are infinitely many zeros ρ of $\zeta(s)$ inside the *critical strip* $0 \le \mathrm{Re}(\rho) \le 1$. These are the *non-trivial zeros.*

21.3. Appreciating Riemann's revolutionary formula

The Riemann Hypothesis proposes that the non-trivial zeros ρ of $\zeta(s)$ all lie on the line $\mathrm{Re}(\rho) = \frac{1}{2}$; that is, they can each be written as $\rho = \frac{1}{2} + i\gamma$. Moreover they come in conjugate pairs $\frac{1}{2} \pm i\gamma$ by exercise 18.6(e), so that the two corresponding terms on the right-hand side of (21.2) contribute

$$\frac{x^{\frac{1}{2}+i\gamma}}{\frac{1}{2}+i\gamma} + \frac{x^{\frac{1}{2}-i\gamma}}{\frac{1}{2}-i\gamma} = x^{\frac{1}{2}} \left(\frac{\cos(\gamma \log x) + 2\gamma \sin(\gamma \log x)}{\frac{1}{4}+\gamma^2} \right) \approx 2x^{\frac{1}{2}} \frac{\sin(\gamma \log x)}{\gamma}$$

when γ gets large (and the error in this approximation is bounded by $x^{1/2}/|\rho|^2$). Therefore

$$\psi(x) \text{ is roughly } x - 2x^{1/2} \sum_{\gamma>0:\ \zeta(\frac{1}{2}+i\gamma)=0} \frac{\sin(\gamma \log x)}{\gamma},$$

with an error bounded by $x^{1/2}\sum_{\rho:\xi(\rho)=0}\frac{1}{|\rho|^2} \ll x^{1/2}$ by exercise 18.10. After partial summation we deduce the approximation

$$\frac{\pi(x) - (\mathrm{Li}(x) - \frac{1}{2}\mathrm{Li}(x^{1/2}))}{\sqrt{x}/\log x} \approx -2 \sum_{\substack{\gamma>0:\\ \zeta(\frac{1}{2}+i\gamma)=0}} \frac{\sin(\gamma\log x)}{\gamma}. \tag{21.4}$$

The error in Riemann's guesstimate is the numerator of the left-hand side.[1] Moreover each of the sine-terms on the right-hand side of (21.4) oscillate and therefore one cannot hope to obtain a better smooth approximation to $\pi(x)$ than Riemann's guesstimate.

The right-hand side of (21.4) is indeed a sum of oscillatory terms, like those one sees in Fourier analysis, and bears much in common with the formula

$$\{x\} - \frac{1}{2} = -2\sum_{n=1}^{\infty}\frac{\sin(2\pi nx)}{2\pi n} \tag{16.1}$$

from chapter 16. The approximation on the right-hand side of (21.4) is also a sum of sine functions, with the numbers γ employed in two different ways in place of $2\pi n$: Each γ is used inside the sine (as the "frequency"), and the reciprocal of each γ forms the coefficient of the sine (as the "amplitude"). We even get the same factor of -2 in each formula. However, the numbers γ in (21.4) are much more subtle than the straightforward numbers $2\pi n$ in (16.1).

We emphasize that (21.4) is valid if and only if the Riemann Hypothesis is true. There is a similar formula if the Riemann Hypothesis is false, but it is complicated and technically unpleasant as the coefficients, $1/\gamma$, are replaced by functions of x. So we believe that the Riemann Hypothesis must hold because it leads to the elegant explicit approximation to the error term in (21.4), and this formula is too beautiful and too compelling to not be true.[2]

Exercise 21.2.* Use (21.3) and partial summation to prove that if the Riemann Hypothesis holds, then

$$\frac{\pi(x) - \mathrm{Li}(x)}{x^{1/2}/\log x} = -1 - \sum_{\substack{\gamma\in\mathbb{R}:\\ \zeta(\frac{1}{2}+i\gamma)=0}} \frac{x^{i\gamma}}{\frac{1}{2}+i\gamma} + O\left(\frac{1}{\log x}\right).$$

21.4. Unconditional estimates for $\pi(x)$

We want to approximate $\psi(x)$ by x, that is, show that $\psi^\flat(x) - x$ is "small". Therefore, by (21.3), we wish to bound

$$\sum_{\substack{\rho:\ \zeta(\rho)=0\\ 0\le\mathrm{Re}(\rho)\le 1}} \frac{x^\rho}{\rho}$$

by $o(x)$, or better. The zeros ρ are complex numbers, say, $\rho = \sigma + i\gamma$, so that

$$x^\rho = x^\sigma(\cos(\gamma\log x) + i\sin(\gamma\log x)).$$

[1]And Gauss's guesstimate in the form $\frac{\pi(x)-\mathrm{Li}(x)}{\sqrt{x}/\log x}$ equals the left-hand side of (21.4) minus $1 + o(1)$.

[2]Indeed the error term in the Prime Number Theorem breaking down into sine functions in (21.4) reminds us of music broken down into its harmonics, and this suggests that the Riemann Hypothesis is true because *the primes have music in them.* (Enrico Bombieri, 1998)

Therefore, as x varies, the ratio x^ρ/x^σ rotates around the unit circle and we might expect that there is significant cancellation over a sum of these. However, we do not have tools to measure that cancellation, so we can only bound our sum by taking the sum of the absolute values of the terms. Now $|\frac{x^\rho}{\rho}| = \frac{x^\sigma}{|\rho|}$ so, as x grows, the quantity that matters the most is the value of $\sigma = \mathrm{Re}(\rho)$. For example if we assume the Riemann Hypothesis, then each $|x^\rho| = x^{1/2}$, and so the sum above is $\leq x^{1/2}\sum_\rho 1/|\rho|$. However, the sum here diverges (which can be deduced from (18.8)) so we are unable to directly apply the beautiful (21.3).

Our plan

In chapter 22 we sketch the usual proof of the Prime Number Theorem by truncating the sum in (21.3). We will also give a more complete proof using a different sum that avoids this difficulty but that yields the Prime Number Theorem without as good an error term: We will show that for the following average of $\psi(x)$,

$$\Psi(x) := \sum_{n\leq x}\Lambda(n)\left(1-\frac{n}{x}\right) = \frac{1}{x}\int_2^x \psi(t)dt,$$

one has $\Psi(x) \sim \frac{x}{2}$ which implies the Prime Number Theorem.

CHAPTER 23

Zeros of $\zeta(s)$ with $\mathrm{Re}(s) = 1$

In this chapter we will prove that there are no zeros of $\zeta(s)$ on the line $\mathrm{Re}(s) = 1$. This was the final step in the proof of the Prime Number Theorem in 1896, and it will be the final step in the proof that we present here.

THEOREM 23.1 (No zeros with $\mathrm{Re}(s) = 1$). *We have $\zeta(1+it) \neq 0$ for all $t \in \mathbb{R}$.*

We will give two proofs by contradiction, so in each we begin by assuming that $\zeta(1 + it) = 0$ for some real $t \neq 0$.

23.1. Sketch of a "hands-on" proof

Our goal is to show that if $\zeta(s)$ has a zero at $s_0 := 1 + it$, then ζ has a pole at $1 + 2it$, contradicting that $\zeta(s)$ is analytic for all $s \neq 1$. Now, since $\zeta(\cdot)$ is analytic at s_0 we can write

$$\zeta(s) = c_r(s - s_0)^r + O((s - s_0)^{r+1}) \quad \text{for some integer } r \geq 1 \text{ with } c_r \neq 0$$

provided $|s - s_0|$ is sufficiently small. Taking $s = s_0 + \frac{1}{\log x}$ for x sufficiently large we have

$$\zeta\left(1 + \frac{1}{\log x} + it\right) \sim \frac{c_r}{(\log x)^r} \quad \text{as } x \to \infty.$$

On the other hand, for large x we will prove in section 23.3 that

$$\left|\zeta\left(1 + \frac{1}{\log x} + it\right)\right| \asymp \left|\prod_{p \leq x}\left(1 - \frac{1}{p^{1+it}}\right)^{-1}\right| \tag{23.1}$$

(that is, one can approximate $\zeta(1+\frac{1}{\log x}+it)$, up to a scalar multiple, by truncating the Euler product for $\zeta(1 + it)$ at $p \leq x$). We compare the two estimates by taking absolute values so that

$$\frac{|c_r|}{(\log x)^r} \asymp \prod_{p \leq x}\left|1 - \frac{1}{p^{1+it}}\right|^{-1} \geq \prod_{p \leq x}\left(1 + \frac{1}{p}\right)^{-1} \geq \prod_{p \leq x}\left(1 - \frac{1}{p}\right) \gg \frac{1}{\log x}$$

(since $|1 - \frac{1}{p^{1+it}}| \leq 1 + |\frac{1}{p^{1+it}}| = 1 + \frac{1}{p}$) by Mertens' Theorem. Letting $x \to \infty$ we deduce that $r \leq 1$, so the only possibility is that $r = 1$. The last displayed equation then becomes

$$\prod_{p \leq x}\left|1 - \frac{1}{p^{1+it}}\right|^{-1} \asymp \prod_{p \leq x}\left(1 + \frac{1}{p}\right)^{-1}.$$

Now $\log|1 - \frac{1}{p^{1+it}}|^{-1} = \frac{\mathrm{Re}(p^{-it})}{p} + O(\frac{1}{p^2})$ as $\log|\cdot| = \mathrm{Re}(\log(\cdot))$. Therefore taking logarithms, reorganizing, and letting $x \to \infty$, we deduce that

$$\sum_p \frac{1 + \mathrm{Re}(p^{it})}{p} \text{ is bounded.} \tag{23.2}$$

Since the numerator of every term is non-negative, this implies that p^{it} is close to -1 for "most" primes p ("most", according to a suitable way of measuring). Squaring suggests that p^{2it} is close to 1 for "most" primes p. More precisely, one deduces from exercise 23.1 that

$$\sum_p \frac{1 - \mathrm{Re}(p^{2it})}{p} \le 4 \sum_p \frac{1 + \mathrm{Re}(p^{it})}{p},$$

so that (23.2) implies that $\sum_p \frac{1-\mathrm{Re}(p^{2it})}{p}$ is also bounded. Threading backwards while modifying the above argument, this estimate and (23.1) yield that

$$\left|\zeta\left(1 + \frac{1}{\log x} + 2it\right)\right| \asymp \prod_{p \le x} \left|1 - \frac{1}{p^{1+2it}}\right|^{-1} \asymp \prod_{p \le x} \left(1 - \frac{1}{p}\right)^{-1} \asymp \log x$$

which implies that $\zeta(s)$ has a pole of order 1 at $s = 1 + 2it$, contradicting the already established result that its only pole lies at $s = 1$.

Exercise 23.1. Use that $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \ge 0$ with $p^{it} = e^{i\theta}$ to deduce that $1 - \mathrm{Re}(p^{2it}) \le 4(1 + \mathrm{Re}(p^{it}))$ for all primes p and reals t.

23.2. A formal proof

We now give another proof which will later be easily modified to other important questions (for example, see the proof of Theorem 28.1):[1] Since $\zeta(1 + it) = 0$ we have $\zeta(1 - it) = 0$ by conjugation. Consider the function

$$A(s) := \zeta(s - it)\zeta(s)^2\zeta(s + it) =: \sum_{n \ge 1} \frac{a_n}{n^s}.$$

The double pole of $\zeta(s)^2$ at $s = 1$ is canceled by the two zeros from $\zeta(s - it)$ and $\zeta(s + it)$. Moreover the poles of $\zeta(s \pm it)$ at $s = 1 \mp it$ are canceled by the corresponding zeros of $\zeta(s)$. Therefore $A(s)$ is analytic in the whole of $\mathbb{C}$, though we will only use that it is analytic on $\{s \in \mathbb{C} : \mathrm{Re}(s) > 0\}$ which we fully proved in section 18.2. It is evident that $A(s) \ne 0$ when $\mathrm{Re}(s) > 1$; otherwise $\zeta(s)$ would have a zero in this half-plane.

For $\mathrm{Re}(s) > 1$ we write the Euler product of $\zeta(s)$ as

$$\log \zeta(s) = \sum_{n \ge 1} \frac{\Lambda(n)}{n^s \log n}$$

so that

$$\begin{aligned}
\log A(s) &= \log \zeta(s - it) + 2\log \zeta(s) + \log \zeta(s + it) \\
&= \sum_{n \ge 1} \frac{\Lambda(n)}{\log n} \left(\frac{1}{n^{s-it}} + \frac{2}{n^s} + \frac{1}{n^{s+it}} \right) \\
&= \sum_{n \ge 1} \frac{\Lambda(n)}{n^s \log n} |1 + n^{it}|^2.
\end{aligned}$$

Therefore the coefficients of $\log A(s)$ are all ≥ 0, and so, exponentiating, $a_n \ge 0$ for all $n \ge 1$. In particular, $a_p = |1 + p^{it}|^2$ and $a_{p^2} = |1 + p^{it} + p^{-it}|^2 + 1 \ge 1$ for every prime p.

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[1] This nice proof was pointed out to me by Sunkai Leung, a student attending this course in 2023, though the idea goes back at least to Narasimhan [**60**] in the 1960s.

The Taylor series for $A(s)$ near to real $\sigma \in (1, \frac{3}{2})$ is given by

$$\sum_{n\geq 1} \frac{a_n}{n^\sigma} n^{\sigma - s} = \sum_{n\geq 1} \frac{a_n}{n^\sigma} \sum_{k\geq 0} \frac{((\sigma - s)\log n)^k}{k!} = \sum_{k\geq 0} c_k (\sigma - s)^k$$

where the $(-1)^k c_k$ are the Taylor series coefficients, with[2]

$$c_k := \frac{1}{k!} \sum_{n\geq 1} \frac{a_n (\log n)^k}{n^\sigma}.$$

Here every $c_k \geq 0$ since each $a_n \geq 0$. This Taylor series has radius of convergence > 1 as $A(s)$ is analytic in $\mathrm{Re}(s) > 0$, by exercise 17.4. Therefore $\sum_{k\geq 0} c_k(\sigma - \frac{1}{2})^k$ converges; that is,

$$A(\tfrac{1}{2}) = \sum_{k\geq 0} c_k (\sigma - \tfrac{1}{2})^k = \sum_{n\geq 1} \frac{a_n}{n^{1/2}} \geq \sum_{p \text{ prime}} \frac{1}{p}$$

as $a_n \geq 0$ for every integer n and $a_{p^2} \geq 1$ for every prime p. This diverges, a contradiction since $A(s)$ is analytic in the whole of $\mathbb{C}$, including at $s = \frac{1}{2}$. □

Exercise 23.2 (Landau's lemma).† Suppose that $A(s) = \sum_{n\geq 1} a_n/n^s$ is a Dirichlet series with each $a_n \geq 0$. Let $\beta := \inf\{\sigma \in \mathbb{R} : \sum_{n\geq 1} a_n/n^\sigma \text{ converges}\}$. Prove that $A(s)$ is analytic for all $s \in \mathbb{C}$ with $\mathrm{Re}(s) > \beta$, but not at $s = \beta$. (Think about how this relates to $\zeta(s)$ and to the proof in this section.)

23.3. Truncating Euler products

We will now prove (23.1). The logarithm of the quotient of the two sides is

$$\log\left(\prod_p \left(1 - \frac{1}{p^{1+\frac{1}{\log x}+it}}\right)^{-1} \cdot \prod_{p\leq x} \left(1 - \frac{1}{p^{1+it}}\right)\right)$$
$$= \sum_{p\leq x} \left(\frac{1}{p^{1+\frac{1}{\log x}+it}} - \frac{1}{p^{1+it}}\right) + \sum_{p>x} \frac{1}{p^{1+\frac{1}{\log x}+it}} + \sum_p O\left(\frac{1}{p^2}\right).$$

The last term is $O(1)$, the middle term is, in absolute value,

$$\leq \sum_{p>x} \frac{1}{p^{1+\frac{1}{\log x}}} = O(1)$$

by exercise 3.10, and the first term is, in absolute value,

$$\leq \sum_{p\leq x} \left| \frac{1}{p^{1+it}} (1 - p^{-\frac{1}{\log x}}) \right| \ll \sum_{p\leq x} \frac{1}{p} \cdot \frac{\log p}{\log x} \ll 1.$$

The logarithm is therefore $O(1)$ and (23.1) follows.

Exercise 23.3.* Let $f(\cdot)$ be a multiplicative function, and suppose there exists a constant $\kappa > 0$ such that $|f(p^r)| \leq \kappa p^{(r-1)/2}$ for all primes p and every $r \geq 1$. Let $F(s) := \sum_{n\geq 1} f(n)/n^s$. Prove that $F(s)$ is absolutely convergent in $\mathrm{Re}(s) > 1$ and that

$$F\left(1 + \frac{1}{\log x} + it\right) \asymp \prod_{p\leq x} \left(1 - \frac{f(p)}{p^{1+it}}\right)^{-1}.$$

[2] We initially assume $\mathrm{Re}(s) > 1$ and s is within a small circle around σ to ensure absolute convergence so we can swap the order of the sums with impunity.

CHAPTER 26

Primes in arithmetic progressions

The integer (a, q) divides every number in the arithmetic progression $a \pmod q$, and so there can be no more than one prime $\equiv a \pmod q$ if $(a, q) > 1$. On the other hand if $(a, q) = 1$, then there appear to be infinitely many primes of the form $qn + a$. For example, if we look at how such primes up to 250 distribute in the different reduced residue classes (mod 10):

$$\begin{aligned}
&11,\ 31,\ 41,\ 61,\ 71,\ 101,\ 131,\ 151,\ 181,\ 191,\ 211,\ 241, \ldots\\
&3,\ 13,\ 23,\ 43,\ 53,\ 73,\ 83,\ 103,\ 113,\ 163,\ 173,\ 193,\ 223,\ 233, \ldots\\
&7,\ 17,\ 37,\ 47,\ 67,\ 97,\ 107,\ 127,\ 137,\ 157,\ 167,\ 197,\ 227, \ldots\\
&19,\ 29,\ 59,\ 79,\ 89,\ 109,\ 139,\ 149,\ 179,\ 199,\ 229,\ 239, \ldots,
\end{aligned}$$

then there seem to be lots of primes in each such arithmetic progression. Moreover there seem to be roughly equal numbers in each, a quarter of the primes up to any given point, and this pattern persists as we look further out. Therefore we expect that for any integer $q \geq 1$, the primes are roughly equally split amongst the reduced residue classes mod q; that is,

$$\#\{\text{Primes } p \leq x : \ p \equiv a \pmod q\} \sim \frac{\pi(x)}{\phi(q)} \quad \text{whenever } (a, q) = 1 \tag{26.1}$$

as $x \to \infty$. This is the analogy for arithmetic progressions of the Prime Number Theorem; we will discuss in the next few chapters the steps that led to the proof of this claim. The first important steps were made by Dirichlet.

26.1. Dirichlet's plan

In section 1.2 we proved Euler's result that there are infinitely many primes in the strong form that $\sum_p 1/p$ diverges. In section 5.5 we surmised that the sum of the reciprocals of the primes $\equiv 1 \pmod 4$ up to x grows at roughly the same speed as the sum of the reciprocals of the primes $\equiv 3 \pmod 4$ up to x. One might therefore guess that the primes (other than the prime divisors of q) are roughly equidistributed among the reduced residue classes mod q, so that if $(a, q) = 1$, then we might have

$$\sum_{\substack{p \text{ prime, } p \leq x \\ p \equiv a \pmod q}} \frac{1}{p} \sim \frac{1}{\phi(q)} \sum_{\substack{p \text{ prime, } p \leq x \\ (p,q)=1}} \frac{1}{p}, \tag{26.2}$$

if x is large, which diverges as $x \to \infty$. The simplest case is when $q = 3$, and we proceed as follows: The following identities distinguish the primes mod 3:

$$\frac{1}{2}\left(1 + \left(\frac{p}{3}\right)\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 3, \\ 0 & \text{if } p \equiv -1 \pmod 3, \end{cases}$$

where $(\frac{\cdot}{3})$ is the Legendre symbol and

$$\frac{1}{2}\left(1-\left(\frac{p}{3}\right)\right)=\begin{cases}1 & \text{if } p\equiv -1 \,(\text{mod } 3),\\ 0 & \text{if } p\equiv 1 \,(\text{mod } 3).\end{cases}$$

Summing up these identities over the primes $p \le x$ yields that

$$\sum_{\substack{p\le x\\ p\equiv \pm 1 \,(\text{mod } 3)}} \frac{1}{p} \;=\; \frac{1}{2}\sum_{p\le x,\ p\ne 3}\frac{1}{p} \pm \frac{1}{2}\sum_{p\le x}\frac{\left(\frac{p}{3}\right)}{p}. \tag{26.3}$$

(We interpret the $\pm$ sign as meaning that the two choices, where $\pm = +$ and where $\pm = -$, are consistent within this expression.) Therefore to deduce a strong form of (26.2) we might aim to prove that

$$\sum_{p\le x}\frac{\left(\frac{p}{3}\right)}{p} \text{ converges to some constant } 2c \text{ as } x\to\infty, \tag{26.4}$$

which would imply that

$$\sum_{\substack{p \text{ prime},\ p\le x\\ p\equiv \pm 1 \,(\text{mod } 3)}} \frac{1}{p} = \frac{1}{2}\sum_{p \text{ prime},\ p\le x}\frac{1}{p} \pm c + o(1) \quad \text{as } x\to\infty,$$

which is significantly stronger than (26.2). The sum in (26.4) converges if and only if the Euler product

$$\prod_p \left(1-\frac{\left(\frac{p}{3}\right)}{p}\right)^{-1}$$

converges. Rewriting each factor of the form $(1-t)^{-1}$ as the geometric progression $1+t+t^2+\cdots$, and then multiplying out the product using the multiplicativity of the Legendre symbol, this equals the Dirichlet series

$$\sum_{n\ge 1}\frac{\left(\frac{n}{3}\right)}{n} \tag{26.5}$$

(providing we can resolve any convergence issues). This converges to a non-zero constant,[1] since it equals

$$\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{7}-\frac{1}{8}\right)+\cdots$$

which is the sum of positive terms and is $\le \frac{1}{1^2}+\frac{1}{4^2}+\frac{1}{7^2}+\cdots<\infty$.

How would this plan generalize? The analogous proof works mod 4 and mod 6 (the other two moduli with $\phi(q)=2$), but as soon as $\phi(q)=4$, say, $q=5$, we need more a complicated formula to distinguish between primes in the four reduced residue classes mod 5. The $\phi(q)$ *Dirichlet characters* $\chi\,(\text{mod } q)$, generalizations of the Legendre symbol which we will define and discuss in chapter 27, provide a way to analogously distinguish between the arithmetic progressions.

[1] At first sight asking that this constant is non-zero seems surprising, but remember that we need to prove that the logarithm of this value converges, and $\log 0$ diverges.

The most difficult issue is to prove that analogy to the sum in (26.5), where each $(\frac{n}{3})$ was replaced by $\chi(n)$, is non-zero. Dirichlet linked this question to a deep problem in algebraic number theory (see section 28.4), a source of inspiration in many of the great developments in number theory ever since.

Exercise 26.1. Prove that (26.2) follows from (26.1) by partial summation.

CHAPTER 34

Distribution of the error term in the Prime Number Theorem

Assume the Riemann Hypothesis throughout this chapter. After partial summation the explicit formula (22.1) gives the (renormalized) error term

$$\frac{\pi(x)-\mathrm{Li}(x)}{\sqrt{x}/\log x} = -1 - \sum_{\substack{\gamma\in\mathbb{R}:\ |\gamma|\le T\\ \zeta(\frac12+i\gamma)=0}} \frac{x^{i\gamma}}{\frac12+i\gamma} + O\left(\frac{1}{\log x}\right) \tag{34.1}$$

in the Prime Number Theorem, writing each $\rho=\frac12+i\gamma$, and $T=x^{1/2}(\log x)^3$.

The approximation in (34.1) suggests that the error term $\pi(x)-\mathrm{Li}(x)$ typically has size $\asymp \frac{\sqrt{x}}{\log x}$ whereas the Gauss-Cramér model in (4.3) suggested the size is typically $\asymp \sqrt{\frac{x}{\log x}}$. Not a huge difference, but enough to make us more wary of the Gauss-Cramér model.

The $x^{i\gamma}$-terms in the sum over zeros in (34.1) slowly rotate around the unit circle as x varies, and so their the real parts average out to 0. Therefore, because of the -1-term, we expect the right-hand side of (34.1) to typically be negative (which is reflected in data for small x) and that is why we graph $\mathrm{Li}(x)-\pi(x)$ rather than $\pi(x)-\mathrm{Li}(x)$ in Figure 34.1.

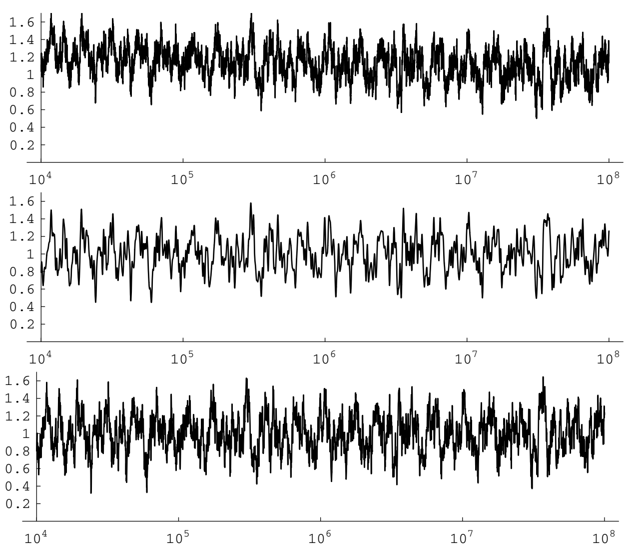

FIGURE 34.1. A graph of $(\mathrm{Li}(x)-\pi(x))/(\sqrt{x}/\log x)$, followed by approximations using the first 100 and 1000 zeros of $\zeta(s)$.

The main question though is how precise an approximation do we get for $\pi(x)-\mathrm{Li}(x)$ by truncating the sum on the right-hand side of (34.1) at the Nth zero of $\zeta(s)$ for various N? Figure 34.1 gives us some idea of how good the approximations are for $N = 100$ and $N = 1000$.

We saw in exercise 16.4 that to obtain a density function for linear combinations of functions like $x^{i\gamma}$ we should use the logarithmic measure and therefore replace x by e^u. We can also pair together the $\frac{1}{2}+i\gamma$ and $\frac{1}{2}-i\gamma$ terms to obtain

$$\Delta(u) := \frac{u(\pi(e^u)-\mathrm{Li}(e^u))}{e^{u/2}} = -1-2\sum_{\substack{\zeta(\frac{1}{2}+i\gamma_n)=0\\ 1\leq n\leq N}} \mathrm{Re}\left(\frac{e^{i\gamma_n u}}{\frac{1}{2}+i\gamma_n}\right)+O\left(\frac{1}{u}\right), \tag{34.2}$$

for $N \geq N(u) := u^4e^{u/2}$, where the zeros of $\zeta(s)$ are $\frac{1}{2}\pm i\gamma_n$ with $0<\gamma_1<\gamma_2<\cdots$.

We wish to determine the value distribution of the right-hand side of (34.2) as u grows. This very much depends on whether the γ_n are linearly independent over $\mathbb{Q}$.

34.1. Linear independence

If γ and γ' are linearly independent over $\mathbb{Q}$, then the values of $\{(\gamma u, \gamma' u) : u \leq U\}$ are uniformly distributed in $(\mathbb{R}/\mathbb{Z})^2$.[1] To see this we note that

$$\begin{aligned}\frac{1}{U}\int_{\substack{0\leq u\leq U\\ \{\gamma u\}\in[\alpha,\beta]\\ \{\gamma' u\}\in[\alpha',\beta']}} 1du &= \frac{1}{U}\int_{u=0}^{U} 1_{\alpha,\beta}(\gamma u)1_{\alpha',\beta'}(\gamma' u)du\\ &= (\beta-\alpha)(\beta'-\alpha')+\sum_{\substack{m,n\in\mathbb{Z}\\ (m,n)\neq(0,0)}} c_m c_n'\cdot\frac{1}{U}\int_{u=0}^{U} e^{2\pi i(m\gamma+n\gamma')u}du\end{aligned}$$

using the Fourier expansion from exercise 16.2 (so that $c_m = \frac{e(-m\alpha)-e(-m\beta)}{2\pi i m}$ and $c_n' = \frac{e(-n\alpha')-e(-n\beta')}{2\pi i n}$). As in exercise 16.3, for any $\epsilon>0$, we may truncate this to a finite sum with $1\leq \max\{|m|,|n|\}\leq M_\epsilon$, and then the key observation is that

$$\frac{1}{U}\int_{u=0}^{U} e^{2\pi i(m\gamma+n\gamma')u}du = \frac{1}{U}\frac{e^{2\pi i(m\gamma+n\gamma')u}}{2\pi i(m\gamma+n\gamma')}\Bigg]_{u=0}^{U} \leq \frac{1}{\pi|m\gamma+n\gamma'|\cdot U} = o_{U\to\infty}(1)$$

since $m\gamma+n\gamma'\neq 0$ as γ and γ' are linearly independent over $\mathbb{Q}$. Summing this up over all such pairs m,n and then letting $\epsilon\to 0$, we deduce that $\{(\gamma u,\gamma' u) : u\leq U\}$ is uniformly distributed in $(\mathbb{R}/\mathbb{Z})^2$.

[1] In other words, $\{(\{\gamma u\},\{\gamma' u\}) : u \leq U\}$ is uniformly distributed in $[0,1)^2$.

Therefore the distribution of values of $\frac{1}{U}\{f(e^{i\gamma u}, e^{i\gamma' u}) : u \le U\}$ is the same as the distribution of values of $\frac{1}{X^2}\{f(e^{ix}, e^{iy}) : x, y \le X\}$, for any polynomial f. This is better described as the same distribution as

$$\{f(\mathbb{X}, \mathbb{Y}) : \mathbb{X}, \mathbb{Y} \text{ uniformly distributed on } \mathbb{U}\}$$

where $\mathbb{U}$ is the unit circle, $\{z : |z| = 1\}$.

Exercise 34.1. We show, with the example $f(x, y) = 4x + y + 3$, that one gets very different distributions if there are dependencies.

(a) Show that if γ and γ' are linearly independent, then the values of $f(e^{i\gamma u}, e^{i\gamma' u})$ are dense in the annulus centered at 3, with boundaries the circles of radii 3 and 5. In particular there exist u-values for which $f(e^{i\gamma u}, e^{i\gamma' u})$ is arbitrarily close to -1.

(b) Show that if $\gamma' = 2\gamma$, then $\mathrm{Re}(f(e^{i\gamma u}, e^{i\gamma' u})) \ge 0$ for all $u \in \mathbb{R}$.

One can determine distributions from moments (see exercise 16.9). Taking the kth moment of the sum on the right-hand side of (34.2) we obtain a sum of terms of the form $\prod_{j=1}^{k}(\frac{1}{2} \pm i\gamma_j)^{-1}$ times

$$\frac{1}{U}\int_0^U e^{i(\pm\gamma_1 \pm \cdots \pm \gamma_k)u}du = \begin{cases} 1 & \text{if } \gamma_1 \pm \cdots \pm \gamma_k = 0, \\ O(\frac{1}{|\gamma_1 \pm \cdots \pm \gamma_k| \cdot U}) & \text{otherwise.} \end{cases}$$

where $\zeta(\frac{1}{2} + i\gamma_j) = 0$ with $0 < \gamma_j \le T$ for each j. The first case happens only when there are linear dependencies amongst the ordinates of the zeros.

Occam's razor suggests *the Linear Independence Hypothesis* (LI), that the non-trivial zeros of $\zeta(s)$, considered with multiplicity, are all $\mathbb{Q}$-linear independent except for the conjugate pairs $\frac{1}{2} \pm i\gamma$; this implies that all zeros are simple (*the Simplicity Hypothesis*). Although there is limited evidence for this Hypothesis[2] we will assume it from here on. If LI holds, then the main terms for all the moments are obtained from pairing up zeros with their conjugates, and so there are no main terms for the odd moments. To prove that these diagonal terms contribute far more than the remaining terms we would need to obtain good bounds on the contribution to the moments of the non-diagonal terms, which is difficult even assuming LI, since many of the sums $|\gamma_1 \pm \cdots \pm \gamma_k|$ might be tiny though not 0.

34.2. Developing a model

If we assume LI, then the moments for the completed sum in (34.2) are the same as the moments of

$$\boxed{-1 + 2\sum_{\substack{\zeta(\rho_n)=0 \\ n \ge 1}} \frac{\mathrm{Re}(\mathbb{X}(n))}{|\rho_n|}} \tag{34.3}$$

where the $\mathbb{X}(n)$ are independent random variables each uniformly distributed on the unit circle and each $\rho_n = \frac{1}{2} + i\gamma_n$.[3] Therefore the distribution of values of the sums in (34.2) and (34.3) are the same.

[2] For example, Best and Trudgian [**6**] showed that there are no solutions to $c_1\gamma_1 + \cdots + c_{500}\gamma_{500} = 0$, where each $c_i \in \mathbb{Z}$ with $|c_i| \le 5.5 \times 10^5$, and $0 < \gamma_1 < \cdots$ are the lowest zeros of $\zeta(s)$.

[3] We can take $|\cdot|$ in the denominators in (34.3) since $-\mathbb{X}/\rho$ and $\mathbb{X}/|\rho|$ have the same distributions.

The equation (34.3) is given by the random variable

$$2\mathbb{Y}-1 \quad \text{where } \mathbb{Y} := \sum_{j\geq 1} \frac{1}{|\rho_j|}\,\mathbb{Y}(j) \text{ with each } \mathbb{Y}(j) = \mathrm{Re}(\mathbb{X}(j)).$$

The $\mathbb{Y}(j)$ are independent random variables with mean 0 and variance $\frac{1}{2}$, and so $\mathbb{Y}$ is a random variable with mean 0 and variance

$$\sigma^2 := \frac{1}{2}\sum_{j\geq 1} \frac{1}{|\rho_j|^2} = \sum_{\rho} \frac{1}{\rho} = .02309570\ldots$$

(assuming the Riemann Hypothesis (RH) here and throughout this chapter). This implies that $2\mathbb{Y}-1$ (that is, (34.3)) has mean -1 and so is biased towards negative values.

Since its variance is bounded but non-zero, $\mathbb{Y}$ cannot be normally distributed (as had been predicted by the Gauss-Cramér model), and so its distribution function must be more complicated. Since $\sum_{n\geq 1} \frac{1}{|\rho_n|}$ diverges (which follows from (18.8)), the range of the function $\mathbb{Y}$ is all of $\mathbb{R}$ (and one can prove that the sum defining $\mathbb{Y}$ converges with probability 1).[4] In particular, $2\mathbb{Y}-1$ must take positive values, which correspond (via RH and LI) to values of x for which $\pi(x) > \mathrm{Li}(x)$. We remarked in chapter 1 that $\pi(x) < \mathrm{Li}(x)$ as far as has been calculated (that is, for all x up to 10^{29}), so if there are x-values for which $\pi(x) > \mathrm{Li}(x)$, then we would like to know how big the first such x is and how often we should expect to find such x-values.

34.3. $\pi(x) > \mathrm{Li}(x)$

The discussion in the last paragraph implies that there are values of u and M for which

$$-1-2\sum_{\substack{\zeta(\frac{1}{2}+i\gamma_m)=0\\ 1\leq m\leq M}} \mathrm{Re}\left(\frac{e^{i\gamma_m u}}{\frac{1}{2}+i\gamma_m}\right) \tag{34.4}$$

is positive. On the other hand if we are given $u = U$ and M for which (34.4) is positive, then, since (34.4) is continuous, there exists $\epsilon > 0$ such that (34.4) is positive for all $u \in [U-\epsilon, U+\epsilon]$. To prove that $\Delta(u) > 0$ (of (34.2) so that $\pi(e^u) > \mathrm{Li}(e^u)$) for some such u-value, we need to show that the contribution of the remaining zeros in the explicit formula, those with larger γ-values, will be negligible. We can do this by averaging over u-values in a suitable way.

Bays and Hudson [**3**] used the first million γ_n to try to identify such a U-value; the first they found was at $U \approx 727.95$. They were then able to use such an averaging argument to prove that $\pi(x) > \mathrm{Li}(x)$ for many x in the vicinity of 1.3982×10^{316}. We believe that Bays and Hudson did correctly identify the smallest

[4]Fix $r \in \mathbb{R}$. For any $\epsilon > 0$ and arbitrarily large integer N we can select signs $\delta_n \in \{-1,1\}$ for which $\sum_{n\leq N} \frac{\delta_n}{|\rho_n|} \in (r-\epsilon, r+\epsilon)$. Suppose that $\mathbb{Y}(n) \in (1-\eta, 1]$ if $\delta_n = 1$ and $\mathbb{Y}(n) \in (-1,-1+\eta]$ if $\delta_n = -1$ for $n \leq N$ where $\eta = \epsilon/\sum_{n\leq N}\frac{1}{|\rho_n|}$ so that $\sum_{n\leq N}\frac{\mathbb{Y}(n)}{|\rho_n|} \in (r-2\epsilon, r+2\epsilon)$. Now if N is sufficiently large, then $\sum_{n>N}\frac{\mathbb{Y}(n)}{|\rho_n|}$ has mean 0 and variance $\leq \epsilon^2$, say and so $|\sum_{n>N}\frac{\mathbb{Y}(n)}{|\rho_n|}| < 2\epsilon$ with probability $\geq \frac{1}{2}$. We deduce that $\mathbb{Y} \in (r-3\epsilon, r+3\epsilon)$ with probability $\geq \frac{1}{2}\prod_{n\leq N} \mathrm{Prob}(\delta_n\mathbb{Y}(n) \in (1, 1-\eta)) > 0$.

x-values for which $\pi(x) > \mathrm{Li}(x)$. However, we cannot be certain, because there might just be a conspiracy of zeta-zeros with much bigger γ-values which yield a smaller x-value with $\pi(x) > \mathrm{Li}(x)$.

34.4. Large error terms

Our model allows us to guess at the maximum error term in the Prime Number Theorem, by understanding the large values of the sum of random variables in (34.3). In [**32**], building on work of Montgomery, it is shown that

$$\textbf{Prob}(\mathbb{Y} > \tau) = \exp(-\{c + o(1)\}\sqrt{\tau}e^{\sqrt{4\pi\tau}}), \tag{34.5}$$

for some constant $c > 0$. From this we predict that

$$\max_{u \le U} \pm\Delta(u) = \{\tfrac{1}{2\pi} + o(1)\}(\log\log U)^2$$

which implies Montgomery's conjecture that

$$\boxed{\max_{X \le x} (\pi(X) - \mathrm{Li}(X)) \sim \frac{1}{2\pi}\frac{\sqrt{x}}{\log x}(\log\log\log x)^2}$$

and

$$\boxed{\min_{X \le x} (\pi(X) - \mathrm{Li}(X)) \sim -\frac{1}{2\pi}\frac{\sqrt{x}}{\log x}(\log\log\log x)^2.}$$

Recently Lamzouri [**49**] showed that Montgomery's conjecture follows from an effective linear independence conjecture.[5]

By selecting U in (34.5) so that the contributions from the first few zeros mostly point in the same direction (that is, the $\arg \gamma U \pmod{2\pi}$ for $\gamma \le \tau$ are roughly equal), and then averaging over u close to U, Littlewood proved unconditionally in 1914 that

$$\max_{X \le x} (\pi(X) - \mathrm{Li}(X)) \gg \frac{\sqrt{x}}{\log x}\log\log\log x$$

and similarly

$$\min_{X \le x} (\pi(X) - \mathrm{Li}(X)) \ll -\frac{\sqrt{x}}{\log x}\log\log\log x.$$

34.5. Rubinstein-Sarnak

Let $\mu(t)$ be the probability that (34.3) is $\le t$, so that

$$\widehat{\mu}(\xi) = \int_{t\in\mathbb{R}} e^{-i\xi t}d\mu(t) = e^{i\xi}\mathbb{E}\left(\prod_{n\ge1} e^{-i\xi\,\mathrm{Re}(c_n\mathbb{X}(n))}\right) = e^{i\xi}\prod_{n\ge1}\mathbb{E}\left(e^{i\xi\,\mathrm{Re}(c_n\mathbb{X}(n))}\right)$$

where $c_n = 2/|\rho_n|$ (changing $-\mathbb{X}(n)$ to $\mathbb{X}(n)$ for convenience). Now we define the Bessel function J_0 of order 0 by

$$J_0(t) := \mathbb{E}(e^{it\,\mathrm{Re}(\mathbb{X})}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{it\cos\theta}d\theta = \sum_{m\ge0}(-1)^m\frac{(t/2)^{2m}}{m!^2}$$

[5] That is, $|\sum_{0<\gamma\le T} c_\gamma\gamma| \gg_\epsilon \exp(-T^{1+\epsilon})$ whenever $c_\gamma \in \mathbb{Z}$, not all zero, with each $|c_\gamma| \le T\log T$.

where $\mathbb{X}$ is a random variable uniformly distributed on the unit circle, so that

$$\widehat{\mu}(\xi) = e^{i\xi} \prod_{n\geq 1} J_0(c_n\xi).$$

Now $c_n = 2/\sqrt{\frac{1}{4}+\gamma_n^2}$, and $\log J_0(t) = -t^2/4 + O(t^4)$ for small t. Therefore

$$\log\left(\prod_{n\geq 1} J_0(c_n\xi)\right) \asymp -\xi^2 \sum_\gamma \frac{1}{\frac{1}{4}+\gamma^2} \asymp -\xi^2,$$

so the infinite product defining $\widehat{\mu}(\xi)$ converges everywhere. Rubinstein and Sarnak [**68**] used this argument to deduce that if the Riemann Hypothesis and the Linear Independence Hypothesis are true, then for any continuous measurable function f we have

$$\lim_{X\to\infty} \frac{1}{\log X}\int_1^X f\left(\frac{\pi(x)-\mathrm{Li}(x)}{x^{1/2}/\log x}\right)\frac{dx}{x} = \int_{t\in\mathbb{R}} f(t)d\mu(t).$$

In particular the probability under this logarithmic measure that $\pi(x) < \mathrm{Li}(x)$ is

$$\lim_{x\to\infty} \frac{1}{\log x}\int_{\substack{0\leq t\leq x\\ \pi(t)<\mathrm{Li}(t)}} \frac{dt}{t} = \int_{v=-\infty}^0 d\mu(v) = \boxed{.99999973\ldots.}$$

That is, although $\pi(x) < \mathrm{Li}(x)$ for 99.999973% of x-values (under the logarithmic measure), just occasionally we have $\pi(x) > \mathrm{Li}(x)$; and so the data in chapter 1, which suggested that $\pi(x) < \mathrm{Li}(x)$ for all x, is misleading.

34.6. Primes in short intervals

In section 34.2 we found that the error term in the Prime Number Theorem *is not* normally distributed, because the variance is bounded (indeed, its value largely depends on $x^{i\gamma_1}$). The distribution behaves differently in short intervals: When counting primes in $(x, x+h]$, a typical term in the explicit formula is

$$\frac{(x+h)^\rho}{\rho} - \frac{x^\rho}{\rho} = \frac{x^\rho}{\rho}\left(\left(1+\frac{h}{x}\right)^\rho - 1\right).$$

This suggests writing $h = \Delta x$ with $\Delta > 0$ "small", so that the explicit formula (22.1) gives

$$\frac{\psi(x+\Delta x)-\psi(x)-\Delta x}{\Delta\sqrt{x}} = -\sum_{\substack{\zeta(\frac{1}{2}+i\gamma)=0\\ |\gamma|\leq T}} c(\gamma)x^{i\gamma} + o(1), \tag{34.6}$$

where T is chosen so that $x^{1/2}(\log x)^2/\Delta = o(T)$ and

$$c(t) := \frac{(1+\Delta)^{\frac{1}{2}+it}-1}{(\frac{1}{2}+it)\Delta} = \begin{cases} 1+O(\Delta|t|) & \text{if } \Delta|t| \ll 1,\\ O(1/\Delta|t|) & \text{for all } |t|.\end{cases}$$

As before, we study the distribution of (34.6) using the logarithmic measure assuming the γ with $\gamma > 0$ are linearly independent, so we model the sum in (34.6) by the sum of random variables

$$\sum_{n\geq 1} c_n\,\mathbb{Y}(n) \tag{34.7}$$

where $c_n = 2|c(\gamma_n)|$, with the $\mathbb{Y}(n)$ as in section 34.2. The mean value of this sum is 0 and its variance is

$$\sigma(\Delta)^2 := 2\sum_{j=1}^{k} |c(\gamma_j)|^2 = (1/\Delta)(\log(1/\Delta) + (1-\gamma-\log 2\pi) + o(1))$$

(which is a challenging calculation). The main contribution to the variance (and therefore to the error term most of the time) comes from the zeros that are at height close to x/h $(= 1/\Delta)$, not from the zeros at a low height.

Since Δ is small, the variance is much bigger than the size of any one term in the sum (which are all $O(1)$), and so we can deduce that our sum is normally distributed, $N(0, \sigma(\Delta))$ (as in exercise 34.2). That is, the assumption of linear independence implies that $\psi(x+y) - \psi(x)$ over $x < X$ is normally distributed with mean y and variance

$$y(\log(X/y) + 1 - \gamma - \log 2\pi + o(1)). \tag{34.8}$$

We have not been precise about how large $\Delta = x/y$ needs to be, but typically y is not much smaller than a constant times x.

Surprisingly this same normal distribution for $\psi(x+y) - \psi(x)$, with the same mean and the same variance, was established by Montgomery and Soundararajan [**58**] using a uniform version of the prime k-tuplets conjecture,[6] in the complementary range $(\log x)^2 < y \le x^{o(1)}$. Having two seemingly very different conjectures yield this same consequence, with the same complicated variance, gives one more confidence that this claim is probably correct.[7]

Exercise 34.2.† Suppose that $\mathbb{Y}_1, \mathbb{Y}_2, \dots$ are identically distributed random variables, each of the form $\mathbb{Y} = \mathrm{Re}(X)$, where X is uniformly distributed on the unit circle.

(a) Prove that $\mathbb{E}(\mathbb{Y}^k) = 0$ if k is odd, and $\mathbb{E}(\mathbb{Y}^k) = \frac{1}{2^k}\binom{k}{k/2}$ if k is even.
Let c_1, c_2 be positive constants for which $c_j = 1+O(j/J)$ for $j \le J$ and $c_j \ll J/j$ for $j > J$; and let $\sigma^2 := \sum_{j\ge1} c_j^2 \sim \kappa J$ for some constant $\kappa > 0$.

(b) Prove that $\sum_{j\ge1} c_j^k \ll_k J$ for all fixed $k \ge 1$.
We wish to prove that the random variable $Y := \sum_j c_j \mathbb{Y}_j$ is normally distributed with mean 0 and variance σ^2. As in Billingsley's proof of the Erdős-Kac Theorem (see section 10.1) it suffices to verify that the moments of $\sum_{j\ge1} c_j\mathbb{Y}_j$ match those of the normal distribution.

(c) Since $\mathbb{E}(\mathbb{Y}^k) = \sum_{j_1,\dots,j_k\ge1} c_{j_1}\cdots c_{j_k}\mathbb{E}(\mathbb{Y}_{j_1}\cdots\mathbb{Y}_{j_k})$, show that a term is non-zero if and only if the j_i's "pair up". Deduce that $\mathbb{E}(\mathbb{Y}^k) = 0$ if k is odd.

(d) Let $k = 2m$. Show the contribution of the terms where the j_i's pair up but the values of the different pairs are distinct is

$$\frac{(2m)!}{2^m m!}\sigma^{2m}\left(1 + O\left(\frac{1}{J}\right)\right).$$

(e) Show that the contribution of all other terms is $O_m(J^{m-1})$ and so deduce that

$$\mathbb{E}(\mathbb{Y}^{2m}) = \frac{(2m)!}{2^m m!}\sigma^{2m}\left(1 + O\left(\frac{1}{J}\right)\right).$$

These are indeed the moments of the normal distribution $N(0,\sigma)$ (with a small error term which is $o_{J\to\infty}(1)$).

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[6]See footnote 1 of section 36.2 on page 193.

[7]Montgomery and Soundararajan proved this under the natural density in the range $(\log x)^2 < y \le x^{o(1)}$. The same result but with logarithmic density follows by partial summation.

CHAPTER 40

Primes missing digits

Most integers have many 0's, 1's, 2's, ..., and 9's in their decimal expansion, so integers missing a given digit, or digits, are rare, making them hard to analyze. For example, if $\mathcal{A}$ is the set of integers with only the digits $7, 8$, and 9 in their decimal expansion, then $|\mathcal{A}(10^k)| = 3^k$ (where $\mathcal{A}(x) = \{a \in \mathcal{A} : a \leq x\}$) as there are 3 possibilities for each of the k digits in the decimal expansion.[1]

40.1. Prime values

When we begin to explore, we find the primes

$$7, 79, 89, 97, 787, 797, 877, 887, 977, 997, \ldots$$

in $\mathcal{A}$, which makes it feel plausible that $\mathcal{A}$ contains infinitely many primes. Let $\pi_{\mathcal{A}}(x) = \#\{a \in \mathcal{A}(x) : a \text{ is prime}\}$. We expect that the elements of $\mathcal{A}$ are independently equidistributed modulo every prime p except perhaps for those dividing the base 10: Since the last digit of an element of $\mathcal{A}$ is $7, 8$, or 9, it is coprime with 10 with probability $\frac{2}{3}$, whereas regular integers are coprime with 10 with probability $\frac{1}{2} \cdot \frac{4}{5} = \frac{2}{5}$, and so we guess that

$$\pi_{\mathcal{A}}(x) \sim \frac{2/3}{2/5} \cdot \frac{|\mathcal{A}(x)|}{\log x} = \frac{5}{3} \cdot \frac{|\mathcal{A}(x)|}{\log x}.$$

More generally, if $\mathcal{A}$ is the set of integers which only have digits from a given set $\mathcal{D} \subsetneq \{0, 1, \ldots, q-1\}$ in their base-q expansion, let $\mathcal{D}_q = \{d \in \mathcal{D} : (d, q) = 1\}$, and then we predict that

$$\pi_{\mathcal{A}}(x) \sim \frac{|\mathcal{D}_q|/|\mathcal{D}|}{\phi(q)/q} \cdot \frac{|\mathcal{A}(x)|}{\log x},$$

via the same reasoning. Maynard proved this [**55, 56**] for certain general families of sparse sets $\mathcal{A}$. His most spectacular result [**55**] yields (close to) the above with $q = 10$ and $|\mathcal{D}| = 9$; that is, Maynard proved that there are roughly the expected number of primes that are missing any *one* given digit in their decimal expansion.[2] His methods can't handle sets as sparse as $\mathcal{D} = \{7, 8, 9\}$ with $q = 10$; that is for another day. We will sketch an argument from [**56**] which gives results of this type but only for bases that are significantly larger than 10.

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[1] So $|\mathcal{A}(x)| \asymp x^{\alpha}$ where $\alpha := \frac{\log 3}{\log 10} = 0.4771\ldots$. Note that $|\mathcal{A}(2 \cdot 10^k)| = |\mathcal{A}(10^k)|$ since no element of $\mathcal{A}$ begins with a "1", so $|\mathcal{A}(x)|/x^{\alpha}$ does not tend to a limit.

[2] Here "roughly" means "up to a multiplicative constant" rather than an asymptotic.

40.2. The missing digit problem

Let $\mathcal{A}$ be the set of integers whose digits come from a given set $\mathcal{D} \subsetneq \{0, 1, \ldots, q-1\}$. Our aim is to estimate $\pi_{\mathcal{A}}(N)$ with $N = q^k$ for very large even integers k using the circle method. If the summand $S_{\mathcal{P}}(\theta)S_{\mathcal{A}}(-\theta)$ in (39.2) is large, then $S_{\mathcal{P}}(\theta)$ and $S_{\mathcal{A}}(-\theta)$ must both individually be large. Now Vinogradov's (39.3) implies that $S_{\mathcal{P}}(\theta)$ is only large when θ is near to a rational with a small denominator. The exponential sum $S_{\mathcal{A}}(\theta)$ behaves differently; it is only large when there are many 0's and $(q-1)$'s in the base-q expansion of θ (see section 40.4). The simplest θ that satisfy both criteria take the form $\theta = \frac{i}{q^g}$ for some small g, and the contribution of those with $\theta = \frac{\ell}{q}$ for some integer ℓ in (39.2) is

$$q^{-k}\sum_{\ell=0}^{q-1} S_{\mathcal{P}}\left(\frac{\ell}{q}\right)S_{\mathcal{A}}\left(\frac{-\ell}{q}\right) = q^{-k}\sum_{a\in\mathcal{A}, a\le q^k}\ \sum_{p \text{ prime}, \le q^k}\ \sum_{\ell=0}^{q-1} e\left(\frac{\ell}{q}(p-a)\right)$$
$$= q^{1-k}\sum_{a\in\mathcal{A}, a\le q^k} \pi(q^k; q, a).$$

Now if $a \in \mathcal{A}$ with $(a, q) = 1$ and base-q digit $d \equiv a \pmod q$, then $d \in \mathcal{D}_q$ and $\pi(q^k; q, a) = \pi(q^k; q, d)$. There are $|\mathcal{D}|^{k-1}$ integers $a \in \mathcal{A}, a \le q^k$ with $a \equiv d \pmod q$, and so this sum becomes, using the Prime Number Theorem for arithmetic progressions and as $|\mathcal{A}(q^k)| = |\mathcal{D}|^k$,

$$q^{1-k}\sum_{d\in\mathcal{D}_q} |\mathcal{D}|^{k-1}\pi(q^k; q, d) \sim \frac{q^{1-k}\cdot|\mathcal{A}(q^k)|}{|\mathcal{D}|}\sum_{d\in\mathcal{D}_q}\frac{1}{\phi(q)}\frac{q^k}{\log q^k}$$
$$= \frac{|\mathcal{D}_q|/|\mathcal{D}|}{\phi(q)/q}\cdot\frac{|\mathcal{A}(N)|}{\log N}, \tag{40.1}$$

which is precisely the prediction that we had for $\pi_{\mathcal{A}}(N)$ above. Therefore we wish to show that the contribution of the remaining summands in (39.2) is negligible, significantly smaller than this main term. It therefore suffices to show that

$$\frac{1}{N}\sum_{\substack{0\le j\le N-1\\ \frac{j}{N}\ne\frac{r}{q},\ 0\le r\le q-1}} \left|S_{\mathcal{P}}\left(\frac{j}{N}\right)\right|\cdot\left|S_{\mathcal{A}}\left(\frac{-j}{N}\right)\right| \ll \frac{|\mathcal{A}(N)|}{(\log N)^A} \tag{40.2}$$

for some $A > 1$. We are looking only at the absolute values of the exponential sums $S_{\mathcal{P}}(\frac{j}{N})$ and $S_{\mathcal{A}}(\frac{-j}{N})$, and not trying to detect any surprising identities or cancellations based on angles.

The major arcs are those $\frac{r}{s} \approx \frac{j}{N}$ with $s \le (\log N)^A$. This includes the $\frac{\ell}{q}$ in the argument above since $q < k^A \le (\log N)^A$ when k is sufficiently large. For any other $s \le (\log N)^A$ for which s divides N (so that if prime p divides s, then p divides q), it is not difficult to show that the contributions of $\frac{r}{s}$ that are close to but not equal to some $\frac{j}{N}$ are negligible. Otherwise we again observe, using the Prime Number Theorem for arithmetic progressions, that

$$S_{\mathcal{P}}\left(\frac{r}{s}\right) \sim \frac{\mu(s)}{\phi(s)}\cdot\pi(N).$$

However, if this is non-zero, then $\mu(s) \neq 0$ so s is squarefree and we deduce that s is a divisor of q, so $\frac{r}{s} = \frac{\ell}{q}$ for some integer ℓ, a case we have already dealt with. We will deal with those $s \leq (\log N)^A$ for which s does not divide N, and so has a prime factor not dividing q, in exercise 40.1.

40.3. What makes restricted digit problems tractable?

In the previous chapter we saw that the ternary Goldbach problem is doable but not the binary Goldbach problem, because we cannot rule out the possibility that the contribution of the minor arcs is larger than that of the major arcs. So why are we able to attack restricted digit problems, which are binary problems? The key is that the exponential sums $S_{\mathcal{A}}(\alpha)$ have an extraordinary structure which makes them usually surprisingly small: We can write

$$\mathcal{A}(N) = \Big\{ n = \sum_{i=0}^{k-1} a_i q^i : \text{ Each } a_i \in \mathcal{D} \Big\}.$$

For each such n we have $e(n\theta) = \prod_{i=0}^{k-1} e(a_i q^i \theta)$, and so

$$\begin{aligned} S_{\mathcal{A}}(\theta) &= \sum_{\text{Each } a_i \in \mathcal{D}} \prod_{i=0}^{k-1} e(a_i q^i \theta) = \prod_{i=0}^{k-1} \Big(\sum_{a_i \in \mathcal{D}} e(a_i q^i \theta) \Big) \\ &= |\mathcal{A}(N)| \cdot \prod_{i=0}^{k-1} \frac{F_{\mathcal{D}}(q^i\theta)}{|\mathcal{D}|} \quad \text{where } F_{\mathcal{D}}(\phi) := \sum_{n \in \mathcal{D}} e(n\phi) \end{aligned} \tag{40.3}$$

since $|\mathcal{A}(N)| = |\mathcal{D}|^k$. It is unusual for an exponential sum of interest to be a product of much simpler exponential sums like this. If the exponential sums in the product in (40.3) were evaluated at points that are independent of each other, then we could focus on each i separately and get the best possible results; however, the value of $q^i\theta \mod 1$ can be used to determine $q^{i+1}\theta \mod 1$ and so these are not independent. However, in practice, especially if q is large, they will be independent enough to get some surprisingly strong upper bounds on $|S_{\mathcal{A}}(\theta)|$ for most θ.

40.4. The major arcs for $S_{\mathcal{A}}(\theta)$

Now if $a, a+1 \in \mathcal{D}$, then

$$|e(a\phi) + e((a+1)\phi)|^2 = 2 + 2\cos(2\pi\phi) < 4\exp(-2\|\phi\|^2),$$

where $\|t\| = \min_{n\in\mathbb{Z}} |t-n|$, so that $|e(a\phi)+e((a+1)\phi)| \leq 2\exp(-\|\phi\|^2)$. Therefore, by the triangle inequality,

$$|F_{\mathcal{D}}(\phi)| \leq |\mathcal{D}| - 2 + 2\exp(-\|\phi\|^2) \leq |\mathcal{D}| \exp\Big(- \frac{\|\phi\|^2}{|\mathcal{D}|} \Big),$$

and substituting this into (40.3) gives

$$|S_{\mathcal{A}}(\theta)| \leq |\mathcal{A}(N)| \exp\Big(- \frac{1}{|\mathcal{D}|} \sum_{i=0}^{k-1} \|q^i\theta\|^2 \Big). \tag{40.4}$$

This is significantly smaller than the trivial $|\mathcal{A}(N)|$ unless $\sum_{i=0}^{k-1} \|q^i\theta\|^2$ is "small", which implies that most of the $\|q^i\theta\|$ are small, say, $\ll 1/\sqrt{k}$.

Now $\|\phi\| \ll 1/\sqrt{k}$ if and only if the first $\frac{\log k}{2\log q}$ base-q digits of $\{\phi\}$ are all 0 or are all $q-1$. If this is true for $\phi = q^i\theta$ for almost all i, then most of the base-q digits of θ are either 0 or $q-1$ (as claimed in the first paragraph of section 40.2).

Exercise 40.1 (The final major arcs).[†] Suppose that $1 < s \le (\log N)^A$ with $(r,s) = 1$ and let S be the largest divisor of s that is coprime to q, and assume that $S > 1$.

(a) Prove there exists $j \ll \log k$ for which $q^j \frac{r}{s} = \frac{R}{S}$ for some integer R.
(b) Show that S is the denominator of $q^i \frac{R}{S}$ for all integers i, and deduce that $\|q^i \frac{R}{S}\| \ge \frac{1}{S}$.
(c) Prove that if $\|q^i \frac{R}{S}\| < \frac{1}{2q^\ell}$, then $\|q^{i+j} \frac{R}{S}\| = q^j \|q^i \frac{R}{S}\|$ for all $j \le \ell$.
(d) Let $m = \lfloor \frac{\log S}{\log q} \rfloor$. Deduce that for any i we have $\|q^i \frac{R}{S}\|^2 + \cdots + \|q^{i+m} \frac{R}{S}\|^2 \ge \frac{1}{4q^2}$.
(e) Use (40.4) to show that $|S_{\mathcal{A}}(\frac{r}{s})| \le |\mathcal{A}(N)| \exp(-c_q \frac{k}{\log k})$, where $c_q > 0$ depends only on q.

It now remains to deal with the minor arcs.

40.5. Upper bounds when missing one digit in base q

We will now work only in the case in which we exclude just one digit, b, in base q so that $\mathcal{D} = \{0, 1, \ldots, q-1\} \setminus \{b\}$ and $F_{\mathcal{D}}(\phi) = \frac{e(q\phi)-1}{e(\phi)-1} - e(b\phi)$. Taking absolute values, we have the trivial upper bound $|F_{\mathcal{D}}(\phi)| \le (q-1)^k$. Using the triangle inequality we have

$$|F_{\mathcal{D}}(\phi)| = \left| \frac{e(q\phi)-1}{e(\phi)-1} - e(b\phi) \right| \le 1 + \frac{|e(q\phi)-1|}{|e(\phi)-1|}$$

$$\le 1 + \frac{2}{|e(\phi)-1|} = 1 + \frac{1}{\sin(\pi\|\phi\|)} \le 1 + \frac{1}{2\|\phi\|} \tag{40.5}$$

where $\|t\| = \min_{n\in\mathbb{Z}} |t-n|$, since $\sin(\pi\|t\|) \ge 2\|t\|$.

Now if $\theta = \sum_{j\ge1} \frac{t_{j-1}}{q^j}$ in base q (with the $t_i \in \{0,1,\ldots,q-1\}$), then

$$q^i\theta \mod 1 = \frac{t_i}{q} + \frac{t_{i+1}}{q^2} + \cdots = \frac{t_i + (q^{i+1}\theta \mod 1)}{q},$$

where $(\phi \mod 1) \in [0,1)$, and so $(q^i\theta \bmod 1) \in [\frac{t_i}{q}, \frac{t_i+1}{q})$. This implies that $\|q^i\theta\| \ge \min\{\frac{t_i}{q}, 1 - \frac{t_i+1}{q}\}$ and so, by (40.5),

$$|F_{\mathcal{D}}(q^i\theta)| \le \min\left\{ q-1, 1 + \frac{1}{\min\{\sin(\pi\frac{t_i}{q}), \sin(\pi\frac{q-1-t_i}{q})\}} \right\},$$

and we obtain, in (40.3),

$$|S_{\mathcal{A}}(\theta)| \le \prod_{i=0}^{k-1} \min\left\{ q-1, 1 + \frac{1}{\min\{\sin(\pi\frac{t_i}{q}), \sin(\pi\frac{q-1-t_i}{q})\}} \right\}. \tag{40.6}$$

In particular if, as is typical, $q^{2/3} < t_i < q - q^{2/3}$, then the ith term in (40.6) is $\ll q^{1/3}$, a big improvement over the Parseval bound $\sqrt{q-1}$. Moreover, the t_i are uniformly distributed in $[0, q-1]$ for almost all θ, in which case we get the bound $|S_{\mathcal{A}}(\theta)| \ll (C + o(1))^k$ where $C := \exp(\frac{4}{\pi} L(2, (\frac{-4}{\cdot}))) \approx 3.209912300$, which is much smaller than $q^{k/2}$. This is very different from the distribution of $|S_T(\theta)|$ for a typical set of integers T, since here the set of θ for which $|S_{\mathcal{A}}(\theta)| > q^{k/3}$, say, has tiny measure. We follow Maynard's argument to exploit this.

40.6. Minor arcs. The mean value of $|S_{\mathcal{A}}(\alpha)|$

Since $\{\frac{j}{q^k} : 0 \le j \le q^k - 1\} = \{\sum_{i=1}^{k} \frac{t_i}{q^i} : 0 \le t_1, \dots, t_k \le q-1\}$ we obtain

$$\sum_{j=0}^{q^k-1} \left| S_{\mathcal{A}}\left(\frac{j}{q^k}\right) \right| \le \prod_{i=0}^{k-1} \sum_{t_i=0}^{q-1} \min\left\{ q-1, 1 + \frac{1}{\sin(\pi \frac{\min\{t_i, q-1-t_i\}}{q})} \right\} \tag{40.7}$$

by (40.6).

Exercise 40.2.*†

(a) Prove that

$$\sum_{t=0}^{q-1} \min\left\{ q-1, 1 + \frac{1}{\sin(\pi \frac{\min\{t, q-1-t\}}{q})} \right\} = 3q - 4 + 2 \sum_{1 \le t < \frac{q-1}{2}} \frac{1}{\sin(\pi \frac{t}{q})} + \frac{1_{2|q-1}}{\sin(\pi \frac{q-1}{2q})}.$$

(b) Show that this equals $\frac{2}{\pi} q \log q + O(q)$.

(c) Deduce that this is $\le (q-1)q^{\tau}$ for some $\tau < \frac{1}{5}$ for q sufficiently large.

(d) Deduce that

$$\sum_{j=0}^{N} \left| S_{\mathcal{A}}\left(\frac{j}{N}\right) \right| \le |\mathcal{A}(N)| \cdot N^{\tau}.$$

(In fact one can take $\tau = \frac{1}{5} - 10^{-9}$ for $q \ge 133359$.)

After some work (given in more detail in [**A-6**, chapter 94]) we obtain (40.2) using (39.3) (and a minor variant), together with exercise 40.2(d). The exponent "$< \frac{1}{5}$" in exercise 40.2(c) is critical because of the $N^{4/5}$ in (39.3). Thus using the explicit result claimed at the end of the exercise we have: *For any integer* $b \in \{0, \dots, q-1\}$ *and* $q \ge 133359$ *there are roughly the expected number of primes which have no digit* b *when written in base* q. To work with smaller q, or larger exceptional sets, we need better bounds on $S_{\mathcal{A}}(\frac{j}{N})$ than (40.7). Such better bounds can be obtained by taking more than one digit into account at the ith stage.